User Guide | ICX-FieldHawk Handheld

ICX Series Real-Time Spectrum Analyzer Control Software

Table of Contents

1. Version Management

This edition documents SpecICX-gen3 as of the baseline engine release used across the ICX series. Future revisions will be logged in this table as functionality is added or changed.

Edition	Description	Date
1.0	Initial BNC edition of the SpecICX-gen3 User Guide.	07/16/2026

2. Preface

SpecICX-gen3 supports three interface layouts:

- Workstation instruments: single-column layout (default) or dual-column layout
- Handheld instruments: tablet layout

This guide uses the tablet layout to illustrate the interface, working modes, and operating steps. The panel arrangement differs slightly between layouts, but the underlying operating logic is identical across all of them.

Figure 1: Tablet layout interface

Figure 2: Workstation single-column layout interface

3. Preparation

3.1 Software Compatibility

SpecICX-gen3 is compatible with the full ICX instrument family running the corresponding engine baseline.

3.2 Operating Environment Requirements

Workstation-style ICX instruments require SpecICX-gen3 to be installed on a host computer. Recommended host requirements are listed below; systems below these specifications may still work but haven't been characterized.

Requirement	Detail
Operating System	Windows 11/10/8/7 (requires VS2019 C++ redistributables); Ubuntu 22.04/20.04/18.04; Debian 12/11/10; Raspberry Pi OS 64-bit
Architecture	Windows: x64, AArch64 (handheld units only); Linux: x64, AArch64
Processor	Windows: Intel i3 or above, or AArch64 (tested on 8CX Gen2); Linux: tested on Raspberry Pi 4B/5, RK3399, RK3588
Memory	4 GB or above
Storage	IQ recording requires sustained write speed above 400 MB/s
Data Interface	USB 2.0 or 3.0 (3.0 recommended); IQ recording bandwidth/duration depends on interface bandwidth
Display Resolution	1280 × 768 pixels or higher

Requirement	Detail
Other	Some antivirus software has been observed to interfere with normal operation

3.3 Default Software Storage Paths

Handheld ICX instruments install SpecICX-gen3 to a default on-device directory; workstation ICX instruments use a user-defined install directory. Within the software directory:

- data folder: recording files, configuration files, and spectrogram CSV data
- images folder: chart image exports
- reports folder: CSV chart data and the matching configuration file

Aside from quick-record and quick-screenshot files, all other recordings and images can be redirected to a user-defined path. Handheld units require an external storage device for a custom path; workstation units can set the path directly.

3.4 Software Acquisition

See the Software and Firmware Update chapter for how to obtain and install the current release. By default, Windows x64, Linux x86_64, and Linux AArch64 builds are provided; contact BNC support if a Windows x86 build is needed.

4. Working Modes Overview

SpecICX-gen3 offers the following working modes: Standard Spectrum Analysis (SWP), IQ Streaming (IQS), Power Detection (DET), Real-Time Spectrum Analysis (RTA), Phase Noise Measurement, Digital Demodulation (option), MSCAN, and Mapping. Each mode's available measurement functions are summarized below.

4.1 Standard Spectrum Analysis

Sweeps across the configured frequency range for frequency-domain trace measurement and analysis. Available functions:

- Panoramic spectrum sweep
- Local spectrum zoom
- Spectrogram
- Recording and playback
- SEM
- Peak and signal tracking
- IM3, Channel Power, OBW, ACPR

- Amplitude correction, Peak table

4.2 Receiver/IQ Streaming Mode

Captures time-domain data within the analysis bandwidth on a specified trigger and returns it for further processing — suited to time-domain recording and basic demodulation analysis. Available functions:

- IQ time-domain waveform, Spectrogram, Power-time waveform
- Multi-channel DDC
- Spectrum analysis, AM/FM demodulation, Audio analysis
- IQ data recording and playback
- Pulse signal detection (option)

4.3 Power Detection Mode

Performs continuous time-domain detection within the analysis bandwidth — suited to observing power-versus-time behavior such as pulse parameter measurement. Available functions:

- Power-time waveform and zoom
- DET data recording and playback
- Pulse signal detection (option)

4.4 Real-Time Spectrum Analysis Mode

Performs real-time spectrum analysis on the time-domain signal, suited to transient and instantaneous signal work such as interference troubleshooting in complex RF environments. Available functions:

- Real-time spectrum probability density plot and spectrogram
- Real-time spectrum data recording and playback

4.5 Harmonic Analysis Mode

Analyzes a signal's harmonics relative to its fundamental, displaying frequency, amplitude, and relative offset for each harmonic — useful for evaluating harmonic distortion and nonlinearity. Available functions:

- Harmonic spectrum plot
- Harmonic measurement table

4.6 Phase Noise Measurement Mode

Provides automated, high-precision phase noise plots and data tables for analyzing carrier phase stability and noise density at various offsets. Available functions:

- SSB phase noise spectrum plot
- Phase noise measurement table

4.7 Digital Demodulation Mode (Option 71)

Demodulates a range of modulated signal types and scores their modulation quality — suited to in-depth quality assessment of known modulation schemes. Available functions:

- Constellation and eye diagram
- Modulated signal spectrum analysis
- Bit table and demodulation
- ASK/FSK/PSK/QAM demodulation

4.8 MSCAN Mode

Lets users define per-band measurement parameters as a configuration list; once configured, the instrument continuously acquires and displays spectrum data across all listed bands — suited to rapid multi-band preview. Available functions:

- Multi-band spectrum preview
- Spectrum analysis, IQ time-domain waveform
- Configuration list

4.9 Mapping Mode

Combines spectrum measurement data with synchronized GPS coordinates to build a heat map of signal strength, peak power, occupied bandwidth, and interference distribution — suited to radio monitoring, signal surveys, and interference hunting. Available functions:

- Georeferenced map import, GPS integration
- Heatmap generation, Spectrum display
- Signal analysis and automatic measurement
- Measurement results table, Task import/export

4.10 Application Software UI Layout

The SpecICX-gen3 interface consists of: Menu, Graph Display Area, Instrument State, Graph Set Area, Main Setting Area, and Parameter Quick Set.

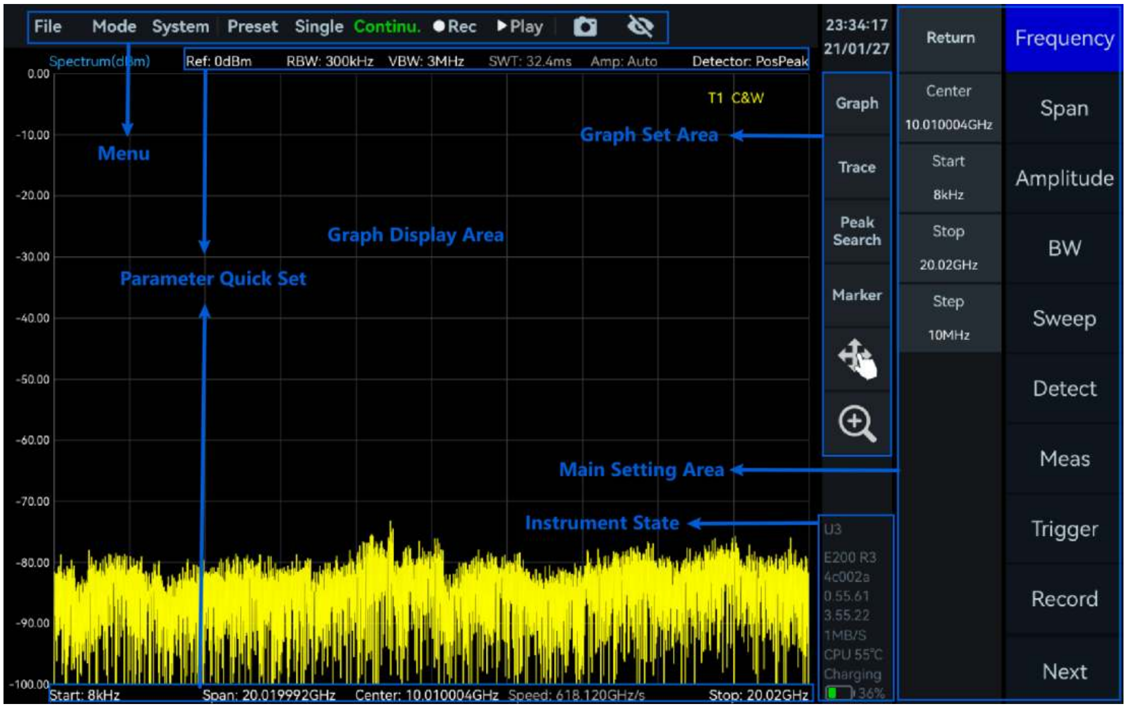

Figure 3: SpecICX-gen3 interface layout

4.10.1 Menu

- Save/load configuration
- Working mode switch
- Single or continuous preview
- Quick screenshot
- Fan control (workstation only)
- Startup state configuration
- Measurement mode selection
- Recording and playback
- GNSS and instrument information view
- Display mode switch (workstation only)

4.10.2 Graph Set Area

- Graph settings, Trace settings
- Marker settings, Measurement results display

4.10.3 Main Settings Area

- Measurement and analysis settings, Trigger settings
- Data recording and playback, System settings

4.10.4 Instrument State

- GNSS antenna connection status, Instrument model
- Last six digits of instrument UID, API/FPGA/MCU version
- Software GUI version, Bus data throughput
- Instrument real-time temperature, Instrument power level (handheld only)

5. Common Operation

5.1 Save and Recall Instrument Configuration

Save the current configuration

1. Click "File" -> "Save State" on the menu bar.
2. In the save-configuration dialog, set the path and file name, then confirm to save.

Open a saved configuration

1. Click "File" -> "Recall State" on the menu bar.
2. Select the configuration file in the file dialog and confirm to load it.

5.2 Save the Pictures

1. Click "File" -> "Save Image" on the menu bar.
2. Set the save path and file name in the dialog, then confirm.
3. A quick-screenshot shortcut button is also available in the menu bar; quick screenshots always save to the /images folder and that path cannot be changed.

5.3 Deleting Files and Images

Handheld ICX instruments: in the relevant storage folder, drag the target file to the trash icon and confirm deletion (the same method applies to recordings and configuration files).

Workstation ICX instruments: select the target file, then use the Edit menu's "Move to Trash" command and confirm.

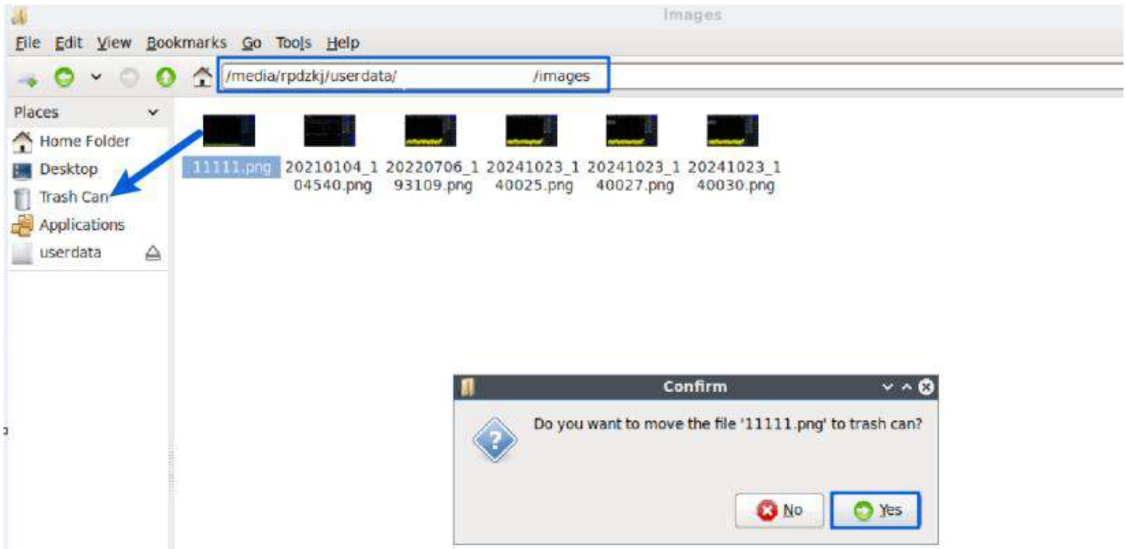

Figure 4: Deleting a picture on a handheld instrument

5.4 Setting the Startup State

The instrument supports three startup behaviors:

Startup State	Behavior
Default	Loads the default configuration.
User Preset	Loads a user-saved profile as the startup configuration.
Last State	Restores the configuration from the previous session.

1. Click "File" -> "Power On State" on the menu bar.
2. Check "Default" or "Last State" directly to use that option on next startup.
3. For "User Preset", choose the saved configuration file in the file dialog; the software will open with that configuration next time.

5.5 Switching Operating Modes

Click "Mode" on the menu bar to switch between Standard Spectrum Analysis, Receiver/IQ Streaming, Power Detection, Real-Time Spectrum Analysis, Digital Demodulation (option), Harmonic Analysis, Phase Noise Measurement, MSCAN, and Mapping.

5.6 Professional and Concise Settings

Click "System" -> "Setting Mode" to switch between "Basic" and "Professional". Professional mode exposes more parameters in the main settings area for users who need finer control.

5.7 Theme Settings

Click "System" -> "Theme" to switch between "Dark" and "Light" display themes.

5.8 Parameter Settings

Click "System" -> "Preference" to access the following settings:

Parameter	Description
Haptics (handheld only)	Vibrates on touchscreen interaction when enabled.
Screen Lock	Shows a lock icon; tap to lock/unlock the screen against accidental input.
Digital Det	Lowers the refresh rate of certain displayed parameters (including markers) to aid observation and recording.
Auto-Dim (handheld only)	Automatically dims the screen after one minute of inactivity.
Brightness (handheld only)	Adjusts screen brightness.
Volume (handheld only)	Adjusts audio volume.
Date/Time (handheld only)	Manual time entry when GNSS is unlocked; one-time or continuous sync to UTC once GNSS locks.
Fan Control	Standard: lower temperature threshold, more responsive. Quiet: higher threshold, smoother ramp.

5.9 GNSS Usage

This section covers acquiring real-time positioning data from the default (factory-installed), EIO (external BNC-supplied GNSS expansion module), and third-party external GNSS sources, plus the 1PPS trigger and 10 MHz reference clock outputs available from GNSS modules.

Parameter	Description
GNSS Source	Default (built-in module); EIO (external BNC GNSS expansion module); External (third-party serial GNSS module).
Baud Rate	Serial baud rate; only needed for an external GNSS source.
Antenna	Internal or external antenna selection (currently external only); needed only with the built-in module.
XPPS Out	On: outputs the XPPS pulse; Off: disables it.
XPPS	Output pulse frequency in Hz. Default 1 Hz (1PPS); 0.5 Hz outputs a pulse every 2 seconds.
Format	Local Time or UTC Time.
SatNum / SNR (Max/Min/Avg)	Number of satellites in view and their signal-to-noise ratios.
Reference Frequency Offset	Offset relative to the 100 MHz reference clock.

5.9.1 Using the Default GNSS Source

1. Connect the GNSS antenna to the device or GNSS module's antenna port.
2. Click "System" -> "GNSS Info" and set "GNSS Source" to "Default".

3. Wait 1-3 minutes for lock; the status-bar GNSS icon turns green once locked, and stays gray otherwise.

Figure 5: Locked GNSS external antenna

5.9.2 1PPS Trigger Using Default GNSS Source

Only IQS, DET, and RTA modes support the GNSS 1PPS trigger. Using IQS mode as the example:

1. Confirm GNSS lock as in 5.9.1.
2. Set "XPPS" to 1 Hz in GNSS Information.
3. Click "Mode" -> "IQStreaming".
4. In the main setup area, click "Next" -> "Trigger" and set "TriggerSource" to "GNSS-1PPS".

Figure 6: Triggering from GNSS 1PPS

5.9.3 Using the 10 MHz Reference Clock from the EIO GNSS Source

Note: Connect the EIO source, instrument, and GNSS antenna before launching the software, or the hardware may not be recognized correctly.

1. Connect the device, EIO source, and external GNSS antenna per the ICX Quick Start Guide.
2. Set "GNSS Source" to EIO and "Antenna" to External; wait 1-3 minutes for lock.
3. Set "DOCXO" to "LockMode" in GNSS Info; wait 5-10 minutes for "DOCXO Locked" to appear.
4. Under System in the main setup area, set "RefCLKSource" to "Internal Clock Source – High Quality" and "RefCLKFreq" to "10 MHz". The reference clock source is now the DOCXO.

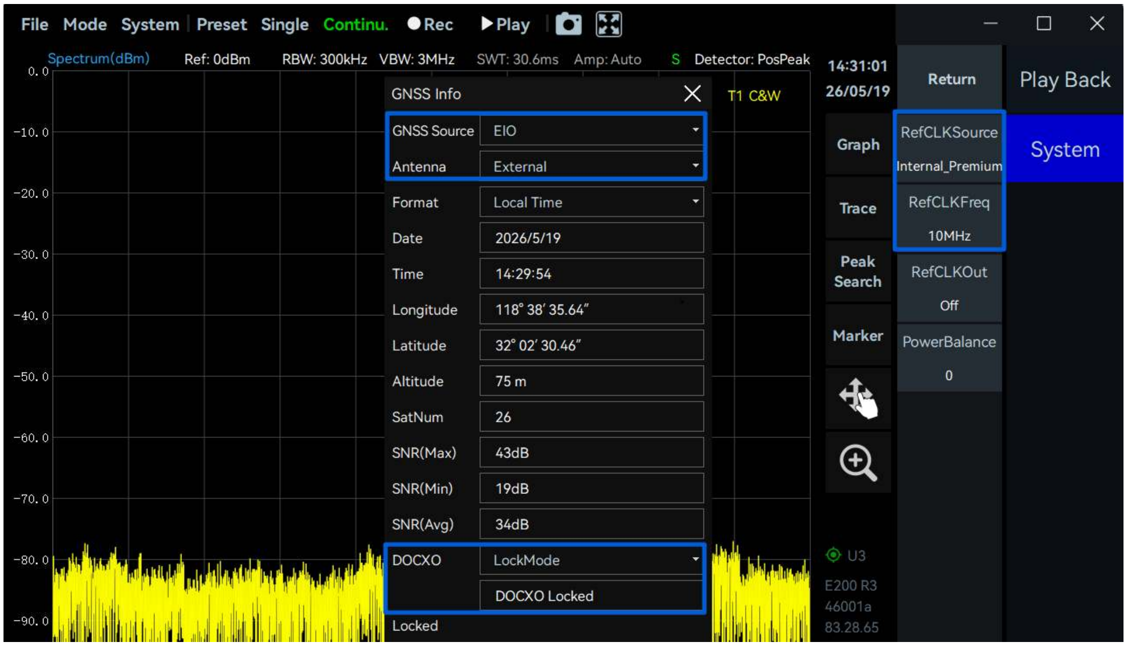

Figure 7: 10 MHz reference clock from a high-quality GNSS module

5.9.4 Using an External GNSS Source

1. Connect the external GNSS module to the host PC's USB port via a USB-to-serial cable.
2. Click "System" -> "GNSS Info".
3. Set "GNSS Source" to "External GPS".
4. Click "Refresh" under "COM Device" and select the newly recognized device.
5. Set "Baud Rate" to match the GNSS module's actual output (e.g. 9600) and click "Connect".
6. The instrument parses and displays the received GNSS data.

Figure 8: Connecting an external GNSS module

5.10 GPIO Output Level Control

With the instrument connected to the matching EIO expansion board, click "System" -> "GPIO" to control output levels per pin ("On" = high, "Off" = low).

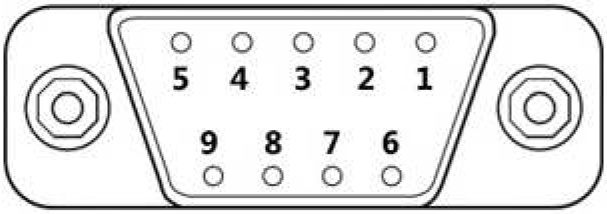

Figure 9: EIO expansion board interface diagram

Pin	Signal
1	GPIO0
2	GPIO2
3	GPIO4
4	GPIO6
5	GND
6	GPIO1
7	GPIO3
8	GPIO5
9	GPIO7

5.11 Viewing Instrument Information

Click "System" -> "About" to view the instrument's UID and firmware version.

5.12 Preset

Click "Preset" on the menu bar to restore the current configuration to the instrument's default state.

5.13 Single or Continuous Preview

Click "Single" for a single-shot preview, or "Continu." for continuous preview.

5.14 Quick Record and Playback

- Recording: click "Rec" to start, "Stop" to end.

- Playback: click "Play" to replay the last recording, "Pause" to pause, and "Continu." to resume live acquisition.

5.15 Hide Panel

Click the hide icon on the menu bar to collapse the main settings panel and expand the display area (tablet layout only).

5.16 Y-Axis Scale Zoom

Click "Graph" -> "Scale/Div" in the chart settings area to change dB-per-division, adjusting the trace's vertical display range. On handheld instruments, the interactive mode button also enables touch gestures: spread two fingers vertically to decrease dB/division, pinch to increase it.

5.17 Offset

1. Click "Graph" -> "Offset" and enter an offset value; positive shifts the trace down, negative shifts it up.
2. Alternatively, enable manual adjustment in the chart settings panel, then press and drag the trace vertically. Restore the default via "Graph" -> "Reset Scale".

5.18 Switch X-Axis Scale

Click "Graph" -> "XScale" to toggle the spectrum's X-axis between Linear and Logarithmic.

5.19 Display Line

Click "Graph" -> "Line" to enable a configurable reference line; set "LinePos" to position it on the Y-axis.

5.20 Setting Chart Units

Click "Graph" -> "Units" to choose the display unit: dBm, dBmV, dBmA, W, V, A, dBuV, dBuA, or dBpW.

5.21 Spectrogram

Only Standard Spectrum Analysis, IQ Streaming, and Real-Time Spectrum Analysis modes support the spectrogram.

Control	Description
Scan Depth	Time length cached on the spectrogram's Y-axis (cache limit: 8000 pixel lines).
Time Density	Spectrogram refresh rate — e.g. 100 scrolls 100 pixel lines per second.
Color	Color gradient for the spectrogram.

1. Click "Graph" -> "Spectrogram" to create a spectrogram alongside the spectrum.
2. Select the spectrogram and click "Graph" to open its settings.
3. "Export Image" saves a PNG to /images; "DataExport" saves cached data (up to the scan depth) as reverse-chronological CSV to /data.

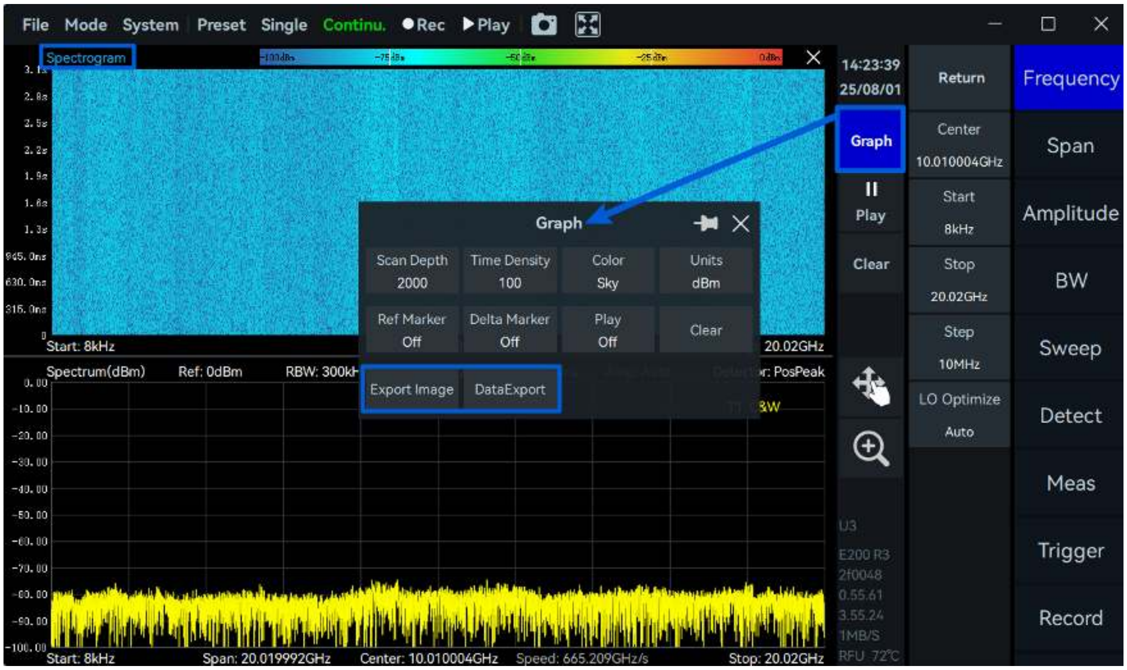

Figure 10: Enabling the spectrogram

4. The spectrogram's color-bar power range follows: Upper limit = Ref.Level + Offset; Lower limit = Ref.Level + Offset − 10 × Scale/Div.

5.22 Export Data

Export the active chart as a PNG image or CSV data via either:

- "Graph" -> "DataExport" in the relevant chart's settings area
- Right-click the chart area and select "DataExport" from the context menu
- "Image" exports a PNG to /images; "Data" exports a CSV to /reports.

5.23 Record and Playback

Record file formats are described in Appendix 1 through Appendix 5.

Parameter	Description
RecordMode – FixedPoints	Preset the point count, duration, and file-size limit; actual duration can't exceed the size limit.
RecordMode – Adaptive	Manually control recording duration (recording still auto-stops if the file exceeds the size limit).

Parameter	Description
RecordTime	Recording duration; applies only in FixedPoints mode.
FileSizeLimit	Storage size limit per recording file.
Disk capacity	Remaining and total disk capacity.
Last / Next frame	Step the playback back or forward one frame.
Back / Forward	Step the playback back or forward multiple frames.

Data Recording

1. Click "Record" -> "RecordMode" in the main setting area and choose the mode.
2. Set "REC File Path" (default /data, or a custom path).
3. In Fixed mode, click "Record" to auto-record a preset size; in Manual mode, click "Record"/"Stop" to control duration directly (recording still stops automatically at the size limit).

Data Playback

1. Click "Play Back" -> "Open File" and select a recording.
2. Click "Playback" to start, "Pause" to pause, "Stop" to exit playback and resume live acquisition; enable "Auto Loop" to loop the file.

5.24 Graph Zoom Function

Two zoom methods are available for examining a local region of a spectrum or time-domain plot: scaling zoom and magnifier zoom.

5.24.1 Zoom Function

Spectrum Zoom (SWP mode only)

1. Click "Graph" -> "Zoom".
2. Select "Spec zoom", then set the frequency range in "Graph", or drag the zoom box and its edges directly on the spectrum.

Figure 11: Spectrum zoom in SWP mode

Time Domain Zoom (IQvT, PvT, and DET modes only)

IQvT/PvT: in IQS mode, open "IQvT" or "PvT", select the channel, enable "Analyze" and "Zoom", then drag the zoom frame or its border to adjust the region.

DET mode: open "Zoom" under "Graph", drag the zoom frame, select "PvT Zoom", and set "TimeCenter"/"TimeRange" under "Graph" to adjust the region.

5.24.2 Magnifying Glass Function

1. Click the magnifying-glass button and frame the area of interest.
2. A thumbnail appears in the upper right showing the full trace and the zoomed region's position.
3. Drag the red box in the thumbnail to reposition, or select a new area.
4. Click the zoom button again to exit and restore the original view.

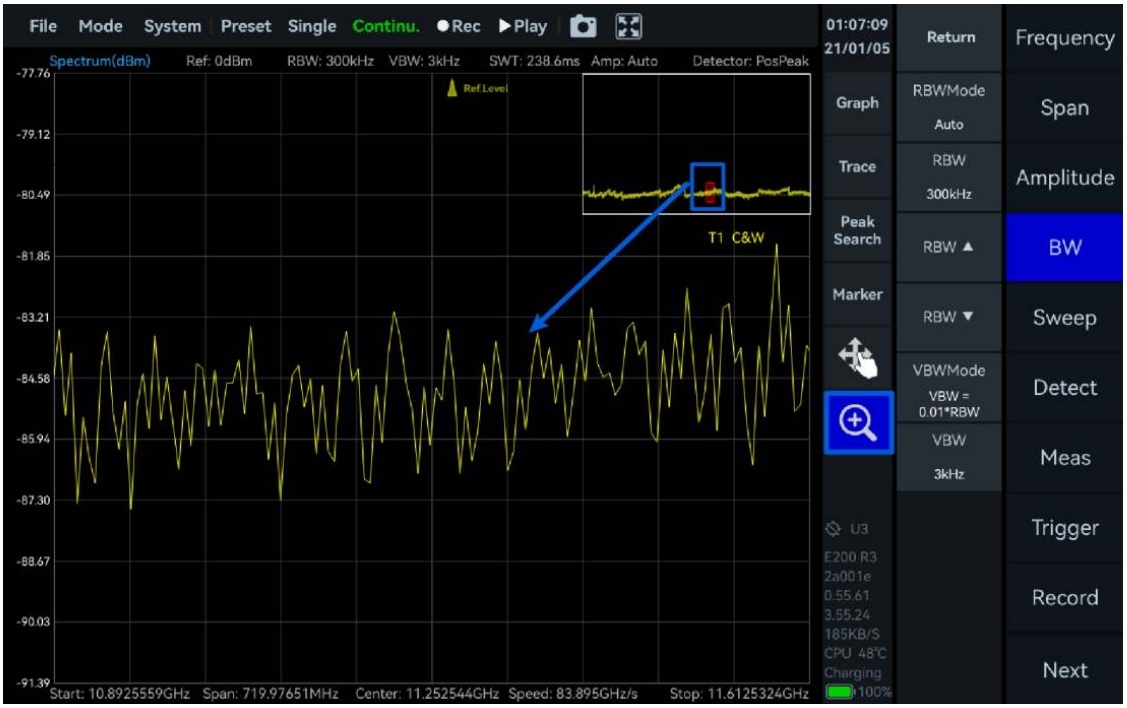

Figure 12: Magnifying-glass zoom in SWP mode

5.24.3 Interactive Zoom (handheld instruments only)

With the interactive mode button enabled, use touch gestures: swipe left/right with one finger to adjust center frequency; spread two fingers horizontally to decrease span; pinch to increase span.

5.25 Trace Settings

Parameter	Description
Enabled	Turns on the trace for a given label (up to 4 traces).
Type	ClearWrite, MaxHold, MinHold, or Average.
Avg	Number of sweeps averaged, for Average-type traces.
StateReset	Clears current trace data and restarts display per the set trace type.
Edit Label	Rename a trace to help manage multiple traces.
Max/Avg/C&W	Enables MaxHold, Average, and ClearWrite traces simultaneously.
Max/Avg/Min	Enables MaxHold, Average, and MinHold traces simultaneously.
Reset	Clears all trace data and reverts to the default ClearWrite type.

5.26 Marker Function

5.26.1 Create Markers

Create a single marker by double-clicking the chart, or via "PeakSearch" in the chart settings area. Create multiple markers via the "Marker" submenu, selecting and enabling each one.

5.26.2 Create Marker Pairs

A marker pair (reference + delta marker, up to 5 pairs) can be created by:

- Enabling a marker pair (e.g. M1R and M1D) in the Marker submenu
- Right-clicking the chart and selecting "Create marker pair"
- Clicking "Graph" -> "Marker Pair" repeatedly to add successive pairs

5.26.3 Close Markers

Close a single marker by disabling it in the Marker submenu. Close all markers via "Graph" -> "Clear All", or by right-clicking the chart and selecting "Clear all markers".

5.26.4 Change Marker Frequency

Enter a value manually: select the marker in the Marker submenu, tap the frequency field, and type the value. Or drag the marker directly, or double-click a target frequency to jump to it.

5.26.5 Marker Switching Traces

Switch a marker's associated trace either from the Marker submenu, or by right-clicking the chart and choosing "Marker trace".

5.26.6 Delta Marker

A delta marker is used with a reference marker to show frequency, time, and amplitude differences relative to that reference.

Figure 13: Enabling a delta marker

5.26.7 Noise Density

After creating a marker, enable "NoiseDensity" in the Marker section to convert displayed power to power density per Hz.

5.26.8 Marker Peak Search

Local peak search: double-click near a local peak, or select the marker and use "Marker" -> "Local Peak". Global peak search: enable "Peak Search" in the graph settings area.

Left/right peak search: with Auto Parameter Setting enabled (default), use "Marker" -> "Left Peak"/"Right Peak" and the software computes thresholds and offsets from the noise floor automatically. For manual control, set "Threshold" and "Excursion" under "Marker" -> "Advanced" first.

Note: Peak excursion is the minimum amplitude difference (dB) between a peak and its adjacent troughs, used to judge prominence.

5.26.9 Marker to Center

Aligns the reference marker's frequency to the chart center, via "Marker" -> "to Center", or the equivalent right-click context menu option.

5.26.10 Marker Switch to Mode

Switches to a different operating mode using the reference marker's current frequency as the new mode's center frequency, via "Marker" -> "Switch to", or the equivalent right-click option.

5.26.11 Signal Tracking

Supported in Standard Spectrum Analysis mode only.

1. Under "Marker" -> "More" -> "Advanced", set "Threshold" and "Shake Range" (jitter within this range won't re-center the frequency).

Figure 14: Setting the peak threshold and shake range

2. Click "Signal Track"; the reference marker searches for peaks within the span and re-centers on the tracked signal as it drifts.

Figure 15: Frequency tracking

Note: This function only shifts the center frequency, not the span; signals that drift beyond the current span may not continue to be tracked, and span narrows further near the instrument's frequency limits.

5.26.12 Peak Tracking

Click "Marker" -> "Peak Tracking" to have the cursor automatically follow the largest peak within the current span in real time.

Note: Peak tracking only operates within the current sweep span.

5.26.13 Peak Table

Supported in Standard Spectrum Analysis mode only.

1. Set the peak threshold under "Marker" -> "More" -> "Advanced" (see Signal Tracking).
2. Click "Peak Table"; the instrument marks and lists up to 10 peaks above the threshold, sorted by descending power, with frequency and power for each.

Figure 16: Peak table

5.27 Quick Parameter Setting

5.27.1 Parameter Setting

Quick-set is supported for Reference Level, RBW, VBW, Detector, Start/Stop Frequency, Span, Center Frequency, and other common parameters — tap the value on screen and type a new one.

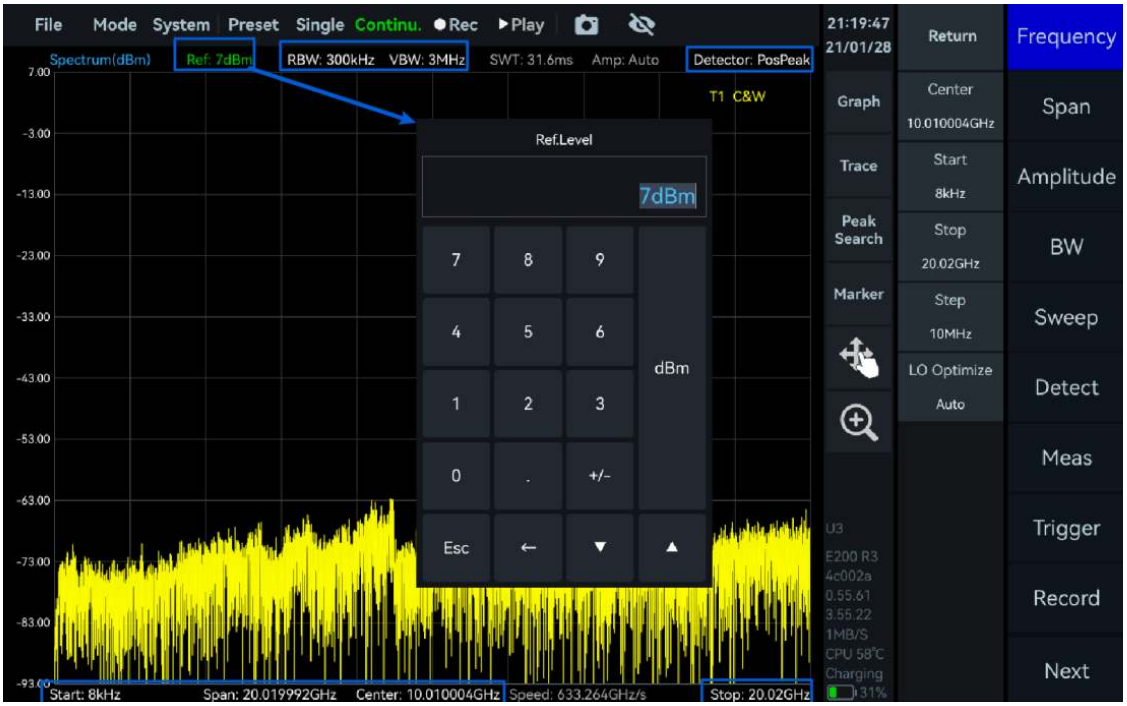

Figure 17: Quick parameter entry

Center frequency and span can also be adjusted via the interactive mode button: drag the trace left/right for center frequency, or scroll the mouse wheel for span.

5.27.2 Parameter Display

- SWT: single-trace sweep time for the current configuration
- Amp: preamplifier status
- Spurious Suppression label: "S" = Standard, "E" = Enhanced, blank = disabled
- Speed: spectrum width swept per unit time (Hz/s)

5.28 Modifying the IQ Sample Rate

In IQ Streaming mode, click "BW" in the main setup area and modify "IQSampleRate" to change the instrument's sampling rate.

5.29 Amplitude Correction

Compensates for external gain/loss via manual entry or an imported frequency-response correction table. Currently supported in Standard Spectrum Analysis mode only.

5.29.1 Correction Rules

- Between the start frequency and the first compensation point, the first point's offset applies.
- Between compensation points, linear interpolation applies.
- Between the last compensation point and the stop frequency, the last point's offset applies.

5.29.2 Amplitude Correction Example

Example: compensate −20 dB from 1-3 GHz, interpolate −20 dB to +10 dB from 3-5 GHz, and compensate +10 dB from 5-7 GHz.

1. Set "Start Frequency" to 1 GHz and "Stop Frequency" to 7 GHz.
2. Click "System" -> "Amplitude Correction".
3. Enable the function, then click "Add" to add a correction entry.
4. Set Frequency 1 = 3 GHz, Offset 1 = −20 dB; add another entry: Frequency 2 = 5 GHz, Offset 2 = 10 dB.
5. Click "Apply" to activate the correction.

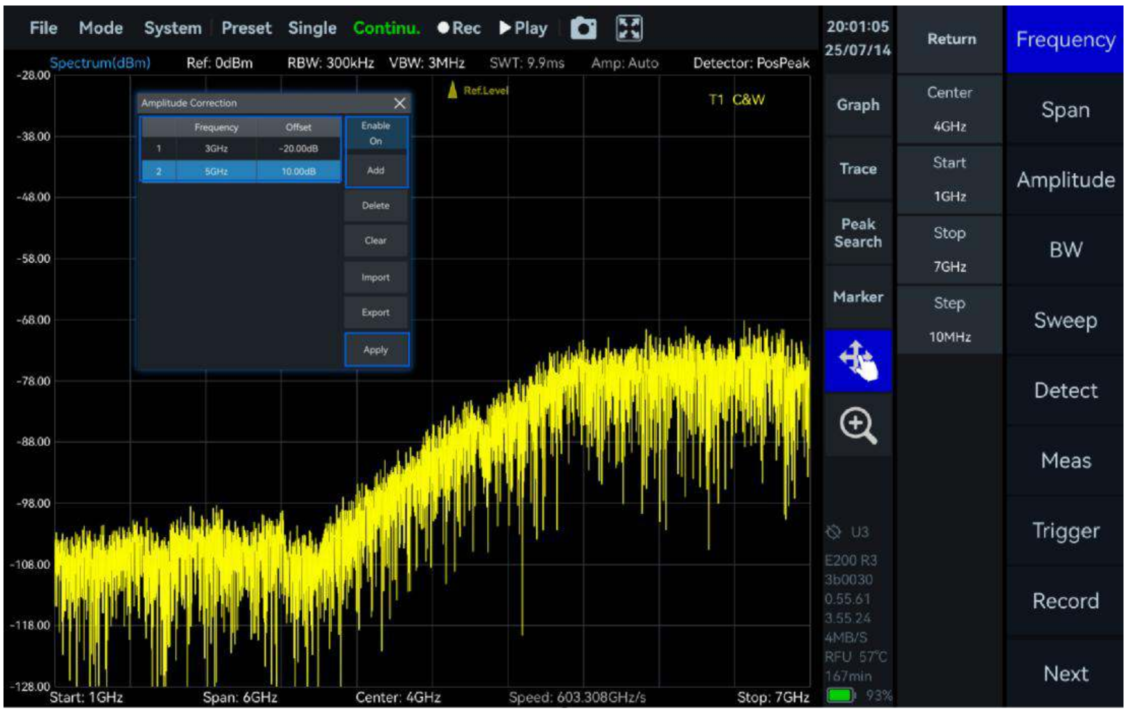

Figure 18: Applying amplitude correction

6. Export the configuration as an Excel file via "Export" (default path /data), or build a custom table in the same format and load it via "Import".

5.30 Antenna Attitude

When an HDA-series handheld directional antenna is connected, SpecICX-gen3 can display the antenna's real-time azimuth, elevation, and polarization angle.

5.30.1 Interface Description

The interface has an attitude-indicator area, an electronic-compass area, and a numeric display area.

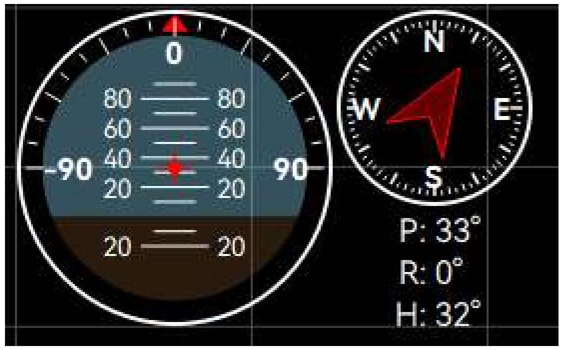

Figure 19: Antenna attitude interface overview

The attitude indicator (left) shows elevation and polarization: blue for upward tilt, brown for downward tilt, with the red arrow at the periphery indicating polarization angle. The compass (right) shows azimuth via a red pointer against North/East/South/West. The numeric area (bottom) shows all three values live.

5.30.2 Parameter Description

Attitude Angle	Description
Elevation	Upward/downward tilt of the antenna's main receiving direction relative to horizontal. Range: -180° to +180°.
Polarization	Rotation of the antenna body around its main receiving-direction axis. Range: -90° to +90°.
Azimuth	Horizontal pointing angle relative to North. Range: 0° to 360°.

5.30.3 Steps for Using with HDA-Series

1. Connect the HDA antenna's communication port to the host via USB Type-C to Type-A, and its RF output to the analyzer's RF input.
2. Start SpecICX-gen3; follow the on-screen calibration prompt to trace a figure-8 with the antenna. The attitude interface then appears automatically (draggable within the app).
3. Hold the antenna normally, pointed at the area under test.
4. The attitude interface updates elevation, polarization, and azimuth in real time.
5. Combine amplitude/field-strength readings with attitude data to help judge the signal source's likely direction.
6. To close the attitude interface, toggle "System" -> "Antenna Attitude"; re-enable via the same menu path.

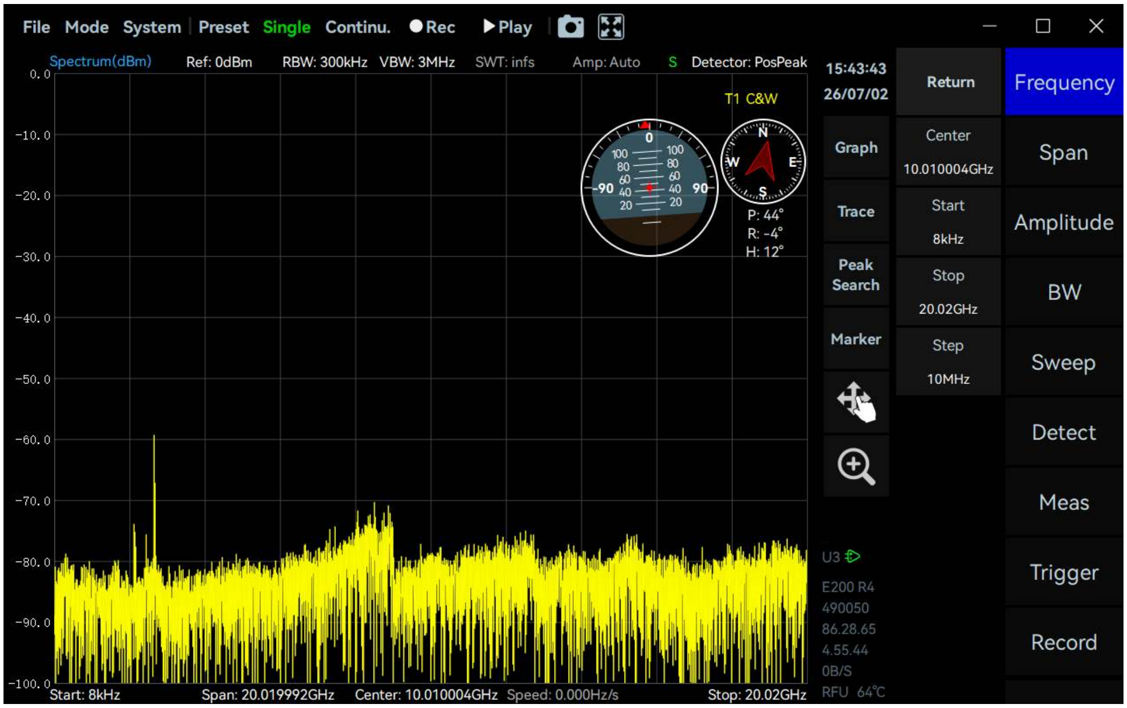

Figure 20: Antenna attitude in use

5.31 Display Modes (workstation instruments only)

Workstation ICX instruments support single-column (default), dual-column, and tablet display modes; handheld instruments support tablet mode only.

Display Mode	Description
Single Column	More spectrum display space, for focused observation.
Dual Column	More convenient parameter access and side-by-side comparison.
Tablet	Clean touch-optimized interface for tablets and handhelds.

5.32 Fan Control (workstation instruments only)

Click "System" -> "Fan Control" to set fan behavior.

Note: Forcing the fan off for extended periods can cause overheating and reduce performance and service life — use with caution.

Fan Setting	Behavior
On	Fan runs continuously.
Off	Fan is disabled.
Auto (default)	Turns on automatically at 50°C and off at 40°C. (Certain fixed-fan models run the fan continuously and are not programmable.)

6. Using the SWP Mode

This chapter covers the key parameters and test methods of Standard Spectrum Analysis (SWP) mode.

6.1 General Parameters of SWP Mode

Parameter	Description
LO Optimize	Auto (default, low spurious); Speed (fast sweep); Spur (low spurious); PhaseNoise (low phase noise).
PreAmplifier	Auto On (engages below approx. −30 dBm ref level); Forced Off; Low/Medium/High Gain.
GainStrategy	LowNoise: flat noise floor, prioritizing low noise. HighLinearity: flat noise floor, prioritizing linearity.
IFGainGrade	0 to X steps, 3 dB per step. Raising the step lowers RF gain and raises the noise floor but improves linearity and reduces spurs; lowering it does the reverse.
Atten	3 dB steps. −1 dB (default) = no attenuation; ≥0 dB enables attenuation.
Amplitude Offset	Shifts the displayed amplitude without altering the underlying signal value.
Auto Reference Level	SWP mode only. Automatically adjusts reference level to keep the trace in a usable display range.
IFAGC	On: reduces IF gain on saturation to prevent overload. Off: disables automatic gain control.
AGC Target	Target power for IF AGC, 0 to −30 dBFS relative to ADC saturation.
AGC Period	−1 s to +2147483 s. Negative: AGC runs once before sampling only. 0: dynamic AGC. Positive: AGC runs on that period.
IF Out	On/off control for the analog IF output.
Antenna Factor	Compensates antenna gain/attenuation to convert received signal into actual field strength.
SWTMode	min SWT and multiples (x2/x4/x10/x20/x50/xN) of the minimum sweep time, or Manual to target a specific sweep time.
PointsStrategy	SweepSpeed: prioritize fastest sweep. PointsAccuracy: prioritize hitting the target point count.
SpurRejection	Bypass, Standard, or Enhanced.
FFTExecution	Auto (CPU below 30 kHz RBW, FPGA above), CPU Preferred, FPGA Preferred, CPU Low/Mid/High Occ, or FPGA only.
Window	FlatTop (amplitude accuracy), B-Nuttall (frequency selectivity), LowSideLobe (low-frequency accuracy), Rectangular (frequency resolution), Kaiser (β = 7.6).
RefCLKSource	Internal; External (falls back to internal if unavailable); External_Forced (requires external clock — no fallback).
RefCLKOut	Outputs a 100 MHz reference clock on current-generation ICX instruments, or 10 MHz on earlier models, when enabled.

6.2 Channel Power

Example signal: BPSK, 1 GHz carrier, −20 dBm, 1 MHz symbol rate.

Parameters

Parameter	Description
Meas BW	Bandwidth of the channel under test; channel power is the integrated power within it.
SpanPower	Sets the measurement bandwidth to the current span and computes channel power over that range.

Procedure

1. Set "Center" to 1 GHz and "Ref.Level" to 0 dBm; click "Meas" -> "ChannelPower".
2. Default parameters populate automatically; channel power, Meas BW, and power density appear in the ChannelPower panel.
3. Adjust channel center frequency and measurement bandwidth by dragging on the chart, or by editing Center/Span/Ref.Level/RBW directly.

Figure 21: Channel power measurement

6.3 Occupied Bandwidth (OBW)

Example signal: BPSK, 1 GHz carrier, −20 dBm, 1 MHz symbol rate.

Parameter	Description
Method	XdB or Percentage.
XdB / %	The specific XdB value or percentage to apply.

Procedure

1. Set "Center" to 1 GHz and "Ref.Level" to 0 dBm; click "Meas" -> "OBW".
2. Click "BW" and set "RBW" to 50 kHz.
3. Default parameters populate; the occupied bandwidth appears in the OBW panel.
4. Adjust Center/Span/Ref.Level/RBW for other signals as needed.

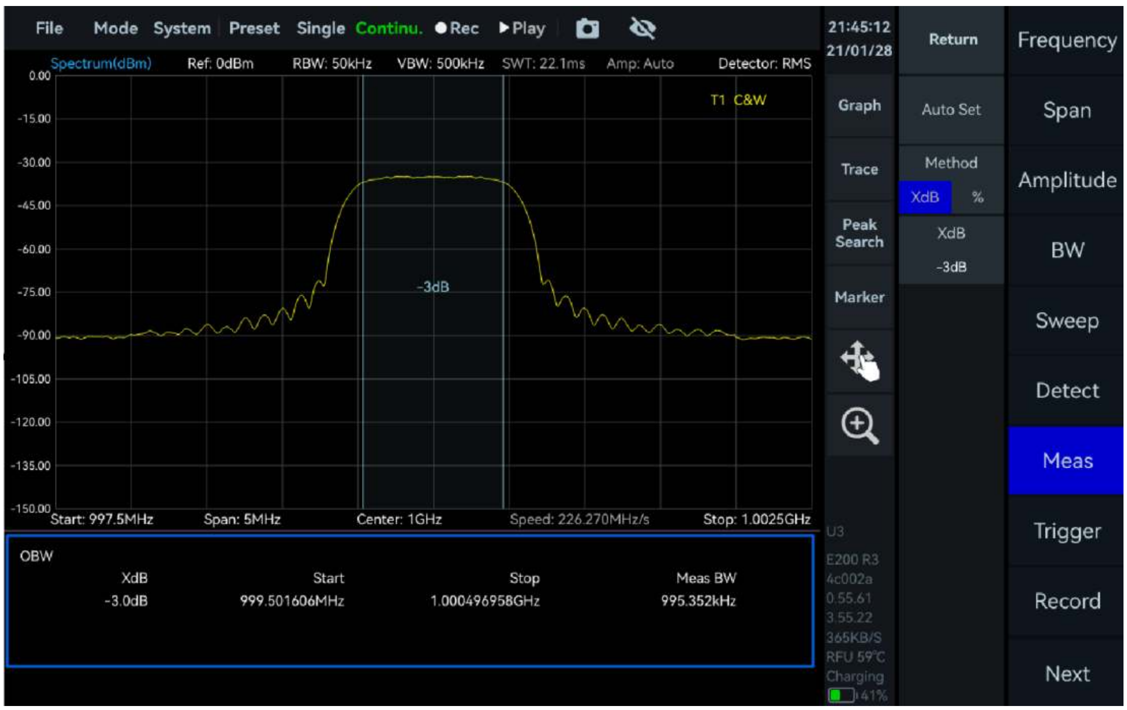

Figure 22: OBW measurement

6.4 Adjacent Channel Power Ratio (ACPR)

Example signal: BPSK, 1 GHz carrier, −20 dBm, 1 MHz symbol rate.

Parameter	Description
Space	Frequency spacing between main and adjacent channels.
Count	Number of adjacent channel pairs, up to 2.

Procedure

1. Set "Center" to 1 GHz and "Ref.Level" to 0 dBm; click "Meas" -> "ACPR".
2. Default parameters populate; view Adj Center, Adj Power, and Adj Ratio in the ACPR panel.

3. Adjust main-channel center frequency, per-channel bandwidth, spacing, and pair count as needed, along with Center/Span/Ref.Level/RBW.

Figure 23: ACPR measurement

6.5 IP3/IM3

Example: IP3/IM3 measurement at 1 GHz.

Parameter	Description
LowTone Frequency/Power	Frequency/power of the injected low-side tone.
HighTone Frequency/Power	Frequency/power of the injected high-side tone.
LowIM3P / HighIM3P Frequency/Power	Frequency/power of the intermodulation products on the low and high sides.
TonePowerDiff	Power difference between the high and low input tones.

Procedure

1. Combine two signals — 999 MHz at 0 dBm and 1.001 GHz at 0 dBm — via two sources and a combiner, then feed the result to the instrument.
2. Set "Center" to 1 GHz and "Ref.Level" to 0 dBm; click "Meas" -> "IM3".
3. Adjust signal power so the third-order intermodulation product sits roughly 6 dB below the reference level.
4. Default parameters populate; view IP3 results in the IM3 panel.

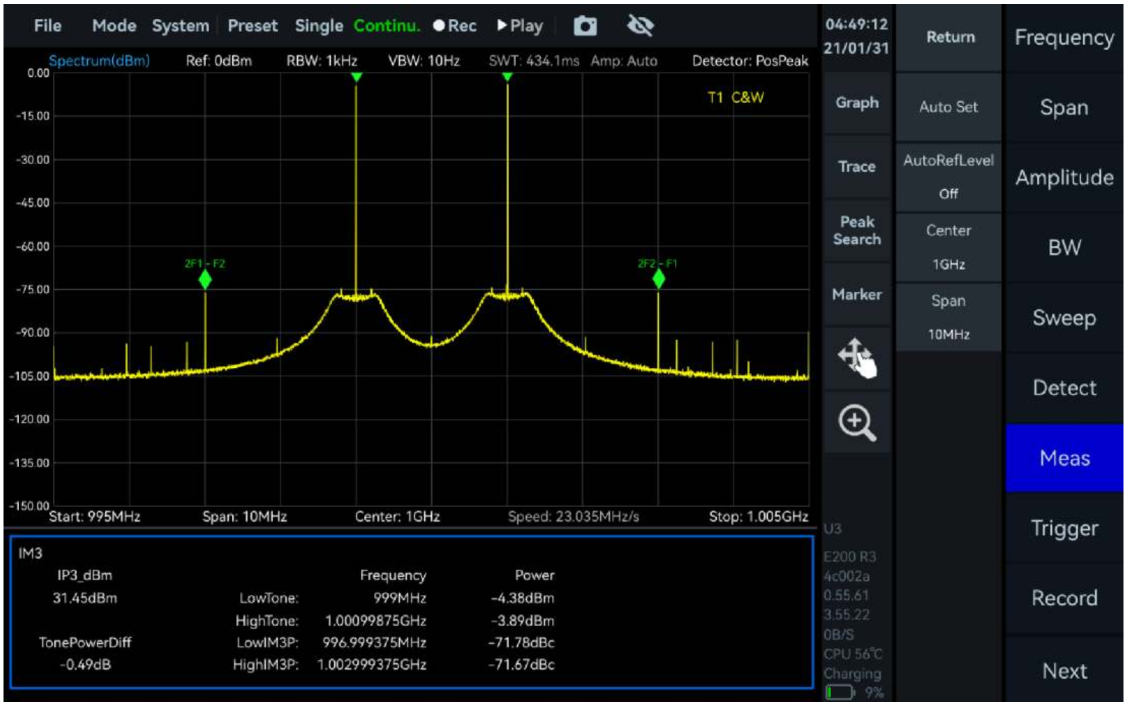

Figure 24: IP3/IM3 measurement

6.6 SEM

Spectrum Emission Mask (SEM) evaluates whether a signal's out-of-band emissions or spurious content stay within limits. Example: an IEEE 802.11ac signal at 1 GHz, −20 dBm.

Parameter	Description
Auto Set	Links to the peak reference type, using the signal peak as the relative reference automatically.
Ref Set Type	Manual (user-defined reference) or Peak (signal peak as reference).
Manual Ref	Reference level value, used only when Ref Set Type = Manual; anchors the offset-table thresholds.
Offset Table – StartFreq/StopFreq	Upper offset band relative to center frequency (system mirrors it below); up to 16 bands.
Offset Table – LimitStart/LimitStop	Power limits for the corresponding offset band.
Offset Table – Mode	Absolute (measured against actual power) or Relative (measured against the reference value).
Offset Table – Priority	Required (must pass) or Suggested (warns if not met).
Save/Load Table	Save or load a measurement template (default path /data).
Load Preset	Loads a built-in template: 802.11a/g, 802.11b, 802.11n (20/40 MHz), 802.11ac (20/40/80/160 MHz), AM NRSC, FM FCC 73.317, FM NRSC Hybrid, AM NRSC 5K/8K Hybrid, Bluetooth.
Export Result	Exports the measurement table (default path /reports).

Procedure

1. Set "Center" to 1 GHz and "Ref.Level" to −20 dBm.
2. Click "Meas" -> "SEM".
3. Click "Offset Table" -> "Load Preset" and choose "802.11ac (20MHz)".
4. Click "BW", set "RBW" to 5 kHz and "VBW Mode" to "VBW = 0.01 RBW".
5. Click "Sweep", set "SWTMode" to "minSWT×20" and "Detector" to "Average".
6. The spectrum plot shows pass/fail against the template, with per-offset-band margins listed below.

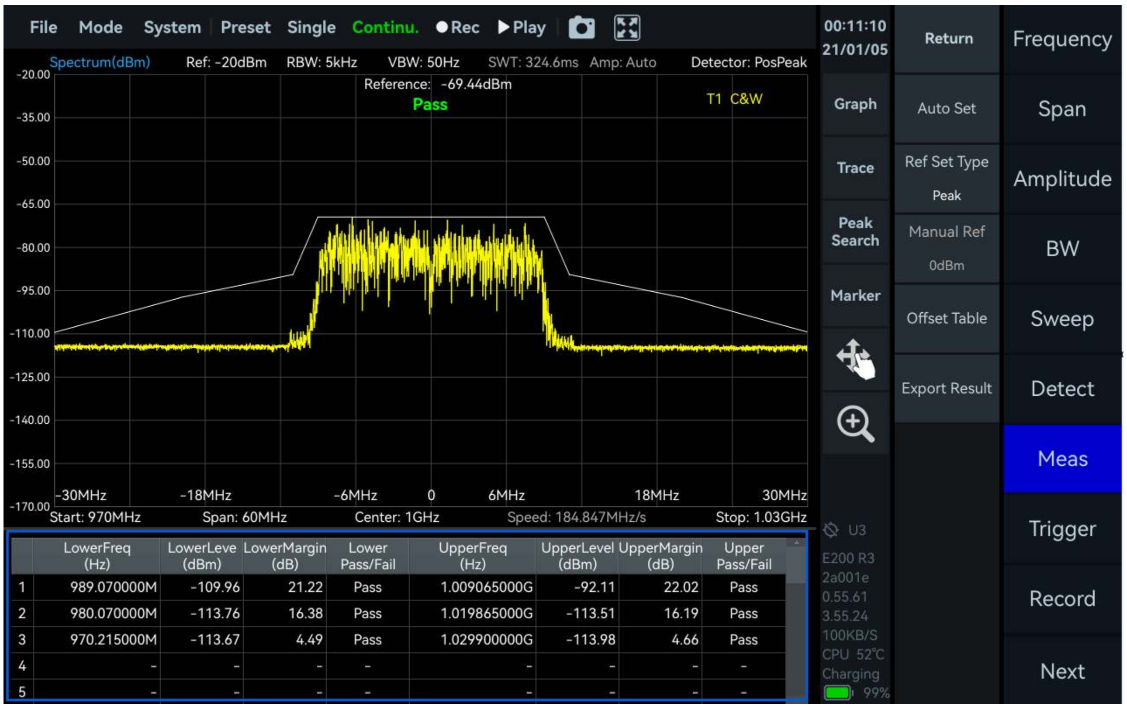

Figure 25: SEM measurement

6.7 Auto Reference Level

Automatically adjusts reference level to keep the trace within a usable range. Example: adjusting for a 1 GHz single-tone signal.

1. Feed a 1 GHz, −35 dBm signal into the instrument.
2. Set "Start Frequency" to 90 MHz and "Stop Frequency" to 3 GHz.
3. Enable "Auto Reference Level" under "Amplitude".
4. Reference level auto-adjusts from 0 dBm to −30 dBm.
5. Raising input power to 5 dBm shifts the reference level from −30 dBm to +10 dBm.

Note: The automatic reference-level floor is −70 dBm, adjusting in 10 dB steps; input should stay within each band's rated loss.

Figure 26: Auto reference level

6.8 Antenna Factor

Compensates antenna gain/attenuation to convert the received signal into actual field strength.

Parameter	Description
Enable	Activates the antenna factor function.
Import	Loads a custom antenna factor table matching the exported file format.
Export	Exports the current configuration as an Excel file (default path /data).
Apply	Applies the current configuration.
Load	One-touch load when paired with an HDA-100 active directional antenna — no manual configuration needed.

Compensation Rules

- From the start frequency to the first compensation point, the first point's factor applies.
- Between compensation points, linear interpolation applies.
- From the last compensation point to the stop frequency, the last point's factor applies.

Configuration Example

Example: 17.5 dB from 30-50 MHz, interpolating to 27.3 dB across 50 MHz-1 GHz, then holding 27.3 dB from 1-2 GHz.

1. Set "Start" to 30 MHz and "Stop" to 2 GHz.
2. Click "Amplitude" -> "Antenna Factor".

3. Enable the function, then click "Add" to add each correction entry.
4. Set Frequency 1 = 50 MHz, Factor 1 = 17.5 dB; add another: Frequency 2 = 1 GHz, Factor 2 = 27.3 dB.
5. Click "Apply".
6. Export the table via "Export", or build one in matching format and load it via "Import".

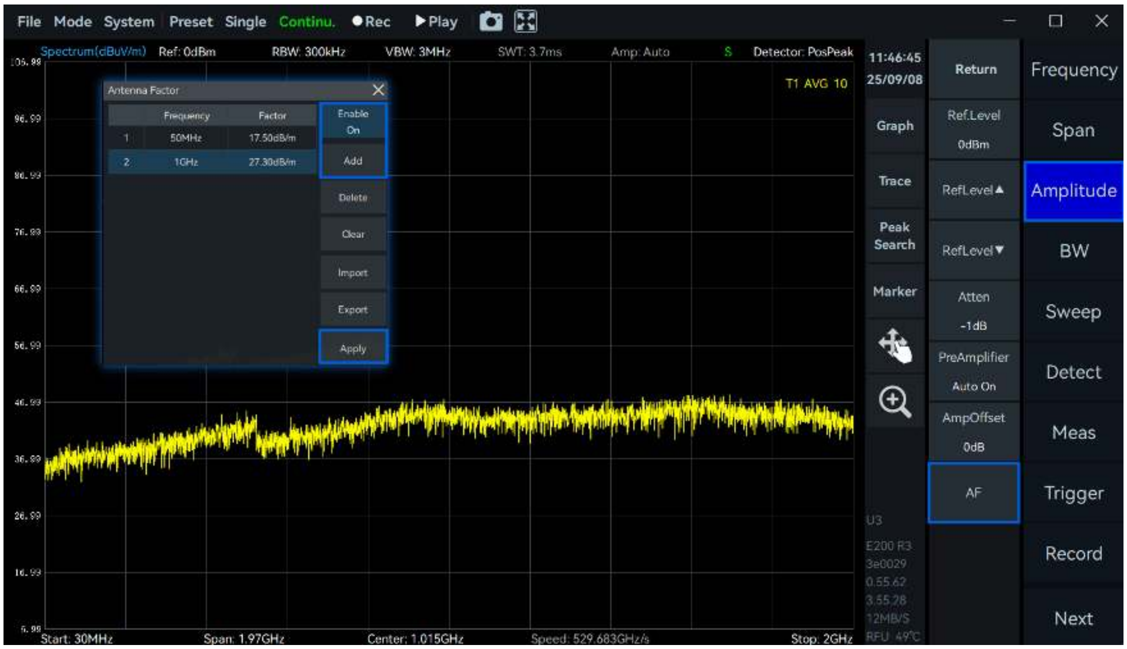

Figure 27: Antenna factor configuration

Auto Load Antenna Factor

1. Connect the instrument and an HDA-100 active directional antenna, then start the software.
2. Set "Start Frequency" to 500 MHz and "Stop Frequency" to 10 GHz.
3. Click "Amplitude" -> "Antenna Factor", enable it, and click "Load" to auto-load the factor.

Figure 28: Automatic antenna factor loading with an HDA-100

7. Using the IQS Mode

This chapter covers the key parameters of IQ Streaming mode and how to further analyze the acquired time-domain IQ data: spectrum analysis, time-domain analysis, power-vs-time analysis, digital down-conversion, and demodulation.

7.1 General Parameters of IQS Mode

Parameter	Description
LO Optimize	See the SWP mode parameter of the same name.
IQSampleRate	ADC sampling rate, 110-130 MSPS.
AnalysisBw	Equivalent sample rate after decimation: Span × 0.8.
Span ▲/▼	Increases/decreases analysis bandwidth in powers of two.
DataFormat	8-bit (lower precision, prone to zeros without signal), 16-bit (default), or 32-bit (higher precision).
PreAmplifier / GainStrategy / IFGainGrade / Atten / AmpOffset / IFAGC / IFAGCTarget / IFOut	See the SWP mode parameters of the same name.
Current IF Gain	Current IF AGC gain in dB; positive = amplification, negative = attenuation.
RecordMode / RecordTime / FileSizeLimit / Disk capacity	See Record and Playback in Common Operation.
RefCLKSource / RefCLKOut	See the SWP mode parameters of the same name.

7.2 IQ Streaming Working Mode Overview

The initial IQS view shows a Max Power vs. Time thumbnail, a spectrum plot, and a time-domain plot. Under "Next" -> "Trigger" in the main setup area, "Preview Time" adjusts the thumbnail's time range. Dragging the "Spectrum-P" and "IQvT-P" selection boxes in the thumbnail lets you inspect different time segments in the spectrum and time-domain views.

7.3 Spectrum Analysis

Parameter	Description
Window	See the SWP mode parameter of the same name.
Intercept	Crops the displayed FFT spectrum — e.g. 0.8 shows 80% of the result, trimming transition-band artifacts.

Procedure

1. Click "FFT" and enable "Analyze"; drag the "Spectrum-P" box in the thumbnail (or set "TimeStart"/"TimeLength" directly) to analyze different time segments. Adjust "Center" and "Span" to change frequency and bandwidth.
2. Use "FFTsize" for point count, "Window" for the window function, "TraceDetector" for the detector, and "Intercept" to crop the transition band.

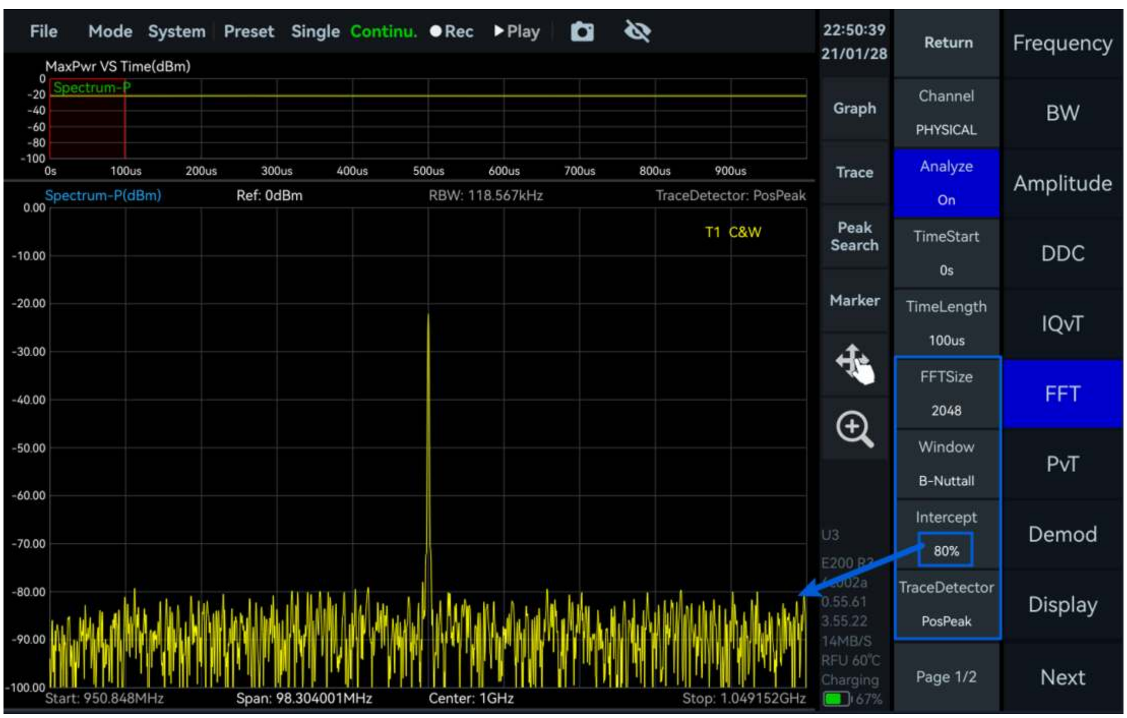

Figure 29: Spectrum analysis of IQ data

7.4 IQvT

1. Click "IQvT", enable "Analyze", and drag the "IQvT-P" box (or set "TimeStart"/"TimeLength"); click "Auto Range" under "Graph" for time-domain analysis of the selected segment.

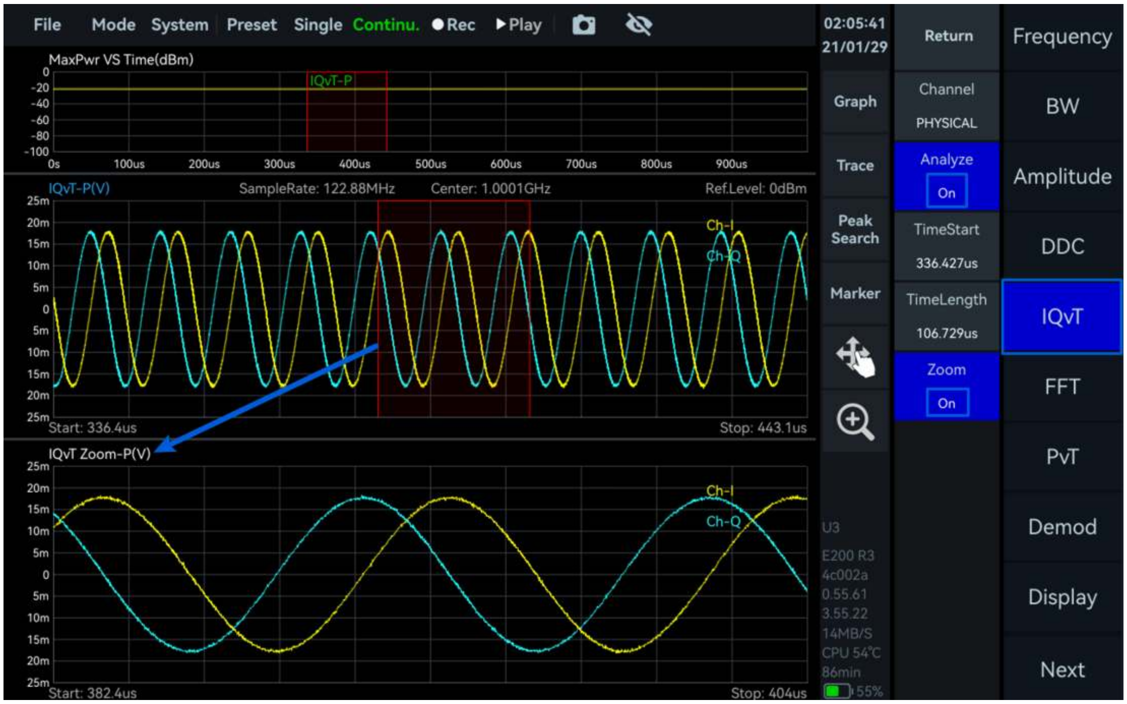

Figure 30: IQ time-domain plot and zoom

7.5 PvT

1. Click "PvT", enable "Analyze", and drag the "PvT-P" box (or set "TimeStart"/"TimeLength") for power-vs-time analysis of the selected segment.

Figure 31: IQ power-time plot and zoom

7.6 AM Demodulation

Example: AM signal, 1 GHz carrier, −20 dBm, 50 kHz modulation rate, 50% modulation depth.

Parameter	Description
n (filter taps)	More taps steepen the filter's transition band and reduce passband ripple.
Fc (cutoff frequency)	0 < Fc < 0.5; e.g. 0.25 low-passes half the bandwidth.
As (stopband attenuation)	Higher values suppress the stopband more strongly, in dB.
mu (fractional sample offset)	Default value recommended.

Procedure

1. Set "Center" to 1 GHz; set "PreviewTime" to 3 ms under "Trigger".
2. Click "Demod" and set "Type" to AM.
3. Expand the analysis range by dragging "Spectrum-P" or increasing "TimeLength" under "FFT".
4. Click "BW" -> "Span▼" to reduce analysis bandwidth (768 kHz in this example).
5. Use the magnifying-glass tool to zoom into the AM waveform; on handheld units, pinch/spread gestures refine the zoom box further.
6. The AM demodulation view shows the modulated spectrum, demodulated waveform, audio spectrum, and parameters including modulation depth, carrier power, modulation rate, SINAD, SNR, and THD.

Figure 32: AM demodulation

Audio Analysis

1. Follow the AM demodulation procedure above.
2. Click "Demod" -> "AudioAnalysis" to enable it, and confirm the measured frequency matches the modulation rate; SINAD and THD are also available.

Figure 33: Audio analysis of AM demodulation

7.7 FM Demodulation

Example: FM signal, 1 GHz carrier, −20 dBm, 5 kHz modulation frequency, 75 kHz frequency offset. Parameters follow the AM demodulation section; low-pass filtering the demodulated signal can reduce high-frequency noise for cleaner audio.

Procedure

1. Set "Center" to 1 GHz; set "PreviewTime" to 3 ms under "Trigger".
2. Click "Demod" and set "Type" to FM.
3. Expand the analysis range as in AM demodulation.
4. Click "BW" -> "Span▼" to set analysis bandwidth (384 kHz in this example).
5. Zoom into the FM waveform using the magnifying-glass tool (plus gestures on handheld units).
6. The FM demodulation view shows modulated spectrum, demodulated waveform, audio spectrum, and parameters including frequency offset, carrier power, frequency error, modulation rate, SINAD, SNR, and THD.

Figure 34: FM demodulation

For audio analysis of the demodulated FM signal, follow the same Audio Analysis steps as AM demodulation.

7.8 DDC Digital Down Conversion

Performs digital down-conversion and resampling on the IQ stream to produce sub-streams for further analysis. Example: single-tone signal at 1 GHz, −20 dBm.

Parameter	Description
OffsetFreq	Complex-mixing frequency shift; positive shifts the spectrum right, negative shifts left.
Decimate	DDC decimation/resampling multiplier.

Procedure

1. Set "Center" to 1 GHz and "Ref.Level" to 0 dBm; select the IQ time-domain plot and choose "Auto Range" under "Graph".
2. Click "DDC", enable Channel 1, and set: Center = 1.003 GHz, OffsetFreq = −3 MHz, Step = 1 MHz, Decimate = 3.
3. Click "FFT", select "DDC1", enable "Analyze", and drag "Spectrum-D1" (or set TimeStart/TimeLength) to analyze the DDC sub-stream.
4. Click "IQvT", select "DDC1", enable "Analyze", and drag "IQvT-D1" similarly.

Figure 35: Time-domain view of a DDC IQ sub-stream

5. Click "PvT", select "DDC1", enable "Analyze", and drag "PvT-D1" similarly.

Figure 36: Power-time view of a DDC IQ sub-stream

8. Using the DET Mode

This chapter covers the key parameters of Power Detection (DET) mode and pulse signal measurement within it.

8.1 General Parameters of DET Mode

Parameter	Description
LO Optimize	See the SWP mode parameter of the same name.
PreAmplifier / GainStrategy / IFGainGrade / Atten / AmpOffset / IFAGC / IFAGCTarget	See the SWP mode parameters of the same name.
IFAGCGain	Current IF AGC gain in dB; positive = gain, negative = attenuation.
RefCLKSource / RefCLKOut	See the SWP mode parameters of the same name.

8.2 Pulse Signal Measurement

Example: pulse-modulated signal, 1 GHz carrier, −10 dBm, 80 µs period, 40 µs width.

Procedure

1. Set "Center" to 1 GHz; click "Single" for single-shot preview.
2. Enable "Zoom" under "Graph" and adjust the zoom region by dragging.

3. On the zoomed plot, create two marker pairs via "Marker Pair": M1R on a pulse's rising edge, M1D on its falling edge, M2R on the same rising edge, M2D on the next pulse's rising edge. M1D then reads pulse width and M2D reads pulse period; duty cycle = M1D ÷ M2D.

Figure 37: Pulse period and width measurement

8.3 Pulse Signal Detection (Option 72)

Requires the Pulse Detection license — see the Application for Options chapter.

Parameter	Description
Threshold	Pulse detection threshold; signal above this level is treated as a valid pulse.
Pulse Count	Maximum number of pulses detected within the current preview time.

Procedure

Example: 1 GHz, −20 dBm pulsed signal, 40 µs width, 80 µs period.

1. Set "Center" to 1 GHz and "Ref.Level" to 0 dBm.
2. Click "BW" and set "AnalysisBW" (61.44 MHz in this example).
3. Click "Trigger" and set "PreviewTime" to 500 µs.
4. Enable "Pulse Det"; drag "Trigger.Level" on the power-vs-time plot to set the threshold, and set "Pulse Count" for the max pulses per preview window.
5. Click "Single"; results show Top/Base Level, Rise/Fall Time and Edge, Width, PRI, and Duty Cycle per pulse, plus Max/Min/Mean PRI and PW with their deviation percentages.

Figure 38: Pulse detection results

9. Using the RTA Mode

This chapter covers key parameters of Real-Time Spectrum Analysis (RTA) mode, disabling the probability density plot, and Wi-Fi signal measurement.

9.1 General Parameters of RTA Mode

Parameter	Description
LO Optimize	See the SWP mode parameter of the same name.
PreAmplifier / GainStrategy / IFGainGrade / Atten / AmpOffset / IFAGC / IFAGCTarget	See the SWP mode parameters of the same name.
IFAGCGain	Current IF AGC gain in dB; positive = gain, negative = attenuation.
SWTMode / Window	See the SWP mode parameters of the same name.
RefCLKSource / RefCLKOut	See the SWP mode parameters of the same name.

9.2 Probability Density Plot

Parameter	Description
BitMap	On/off toggle for the probability density plot.
Color	Sky, Sea (default), Jet, Cold, Hot, or Gray.

Parameter	Description
Afterglow	Increase to extend trace persistence (good for bursts); decrease for a faster refresh (good for continuous signals).

To disable the plot, open "Graph" and turn off "BitMap".

Figure 39: Disabling the probability density plot

9.3 Wi-Fi Signal Measurement

1. Connect the antenna to "RFIN".
2. Set "Center" to 2.44 GHz, and raise "Afterglow" under "Graph" for a clearer view of the Wi-Fi signal.

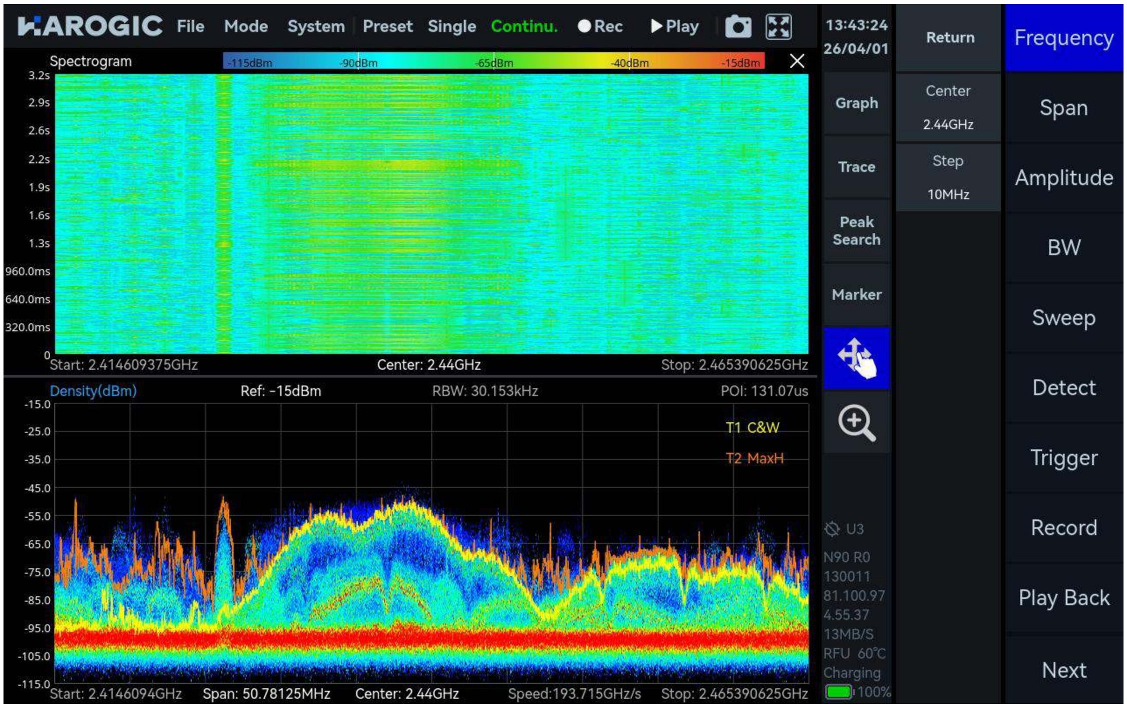

Figure 40: Probability density plot of a Wi-Fi signal

10. Digital Demodulation (Option 71)

Requires the Digital Demodulation license and library — see the Application for Options chapter.

10.1 Parameters

Parameter	Description
SymbolRate	Symbols per second transmitted by the signal; must match the actual modulated signal for correct demodulation.
ModType	2ASK, 2FSK, 4FSK, GMSK, BPSK, QPSK, 8PSK, 16QAM, 32QAM, 64QAM, 128QAM, 256QAM.
FilterAlpha	Filter roll-off rate; must match the transmitter's roll-off for correct demodulation.
Average number	Higher averaging reduces jitter in parameters such as EVM.

10.2 Function Overview

The initial view shows the modulated signal's spectrum, the demodulated constellation and eye diagrams, and quality parameters: EVM, magnitude error, phase error, frequency error, SNR/MER, and a partial decoded bit sequence.

10.3 Procedure

Example: 64QAM signal, 1 GHz, −20 dBm, 1 MHz symbol rate, 0.35 filter alpha.

1. Click "Mode" -> "Digital Demod".
2. Set "Center" to 1 GHz and "Ref.Level" to 0 dBm.
3. Click "Demod" and set ModType = QAM64, SymbolRate = 1 MSPS, FilterType = 0.35, Average Count = 10, then click "Single".

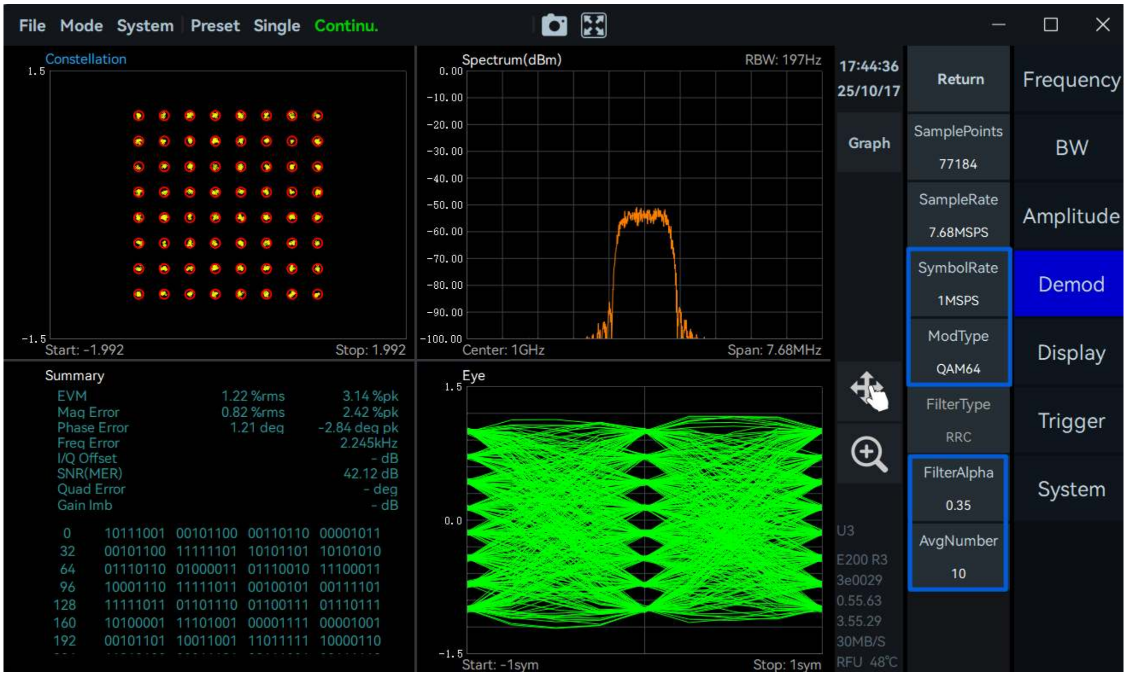

Figure 41: 64QAM demodulation

11. Harmonic Analysis Mode

11.1 Version Requirements

Check the software/firmware version under Viewing Instrument Information, and confirm it meets the minimum GUI, API, FPGA, and MCU versions required for this feature; update per the Software and Firmware Update chapter if needed.

11.2 Enabling Harmonic Analysis

Once the software and firmware meet the version requirement, restart SpecICX-gen3 and click "Mode" -> "Harmonics".

11.3 Parameters

Parameter	Description
Center	Center frequency of the fundamental.

Parameter	Description
Span	Per-harmonic measurement bandwidth, 10 Hz to 100 MHz.
Offset	Shifts the spectrum plot along the amplitude axis.
Harm Count	Number of harmonics measured and plotted, up to 10.
Meas Type	Peak (peak power of fundamental and harmonics) or ChannelPower (channel power within each sweep bandwidth).
Trace Type	ClearWrite (real-time updates) or MaxHold (captures instantaneous peaks).
PK Tracking	Enables peak tracking of the fundamental, centering it automatically.
THD	Evaluates overall signal distortion.

11.4 Procedure

Example: third-harmonic measurement of a 1 GHz, −20 dBm signal.

1. Set "Center" to 1 GHz.
2. Set "Ref.Level" to −10 dBm.
3. Set "Harm Count" to 3 and enable "PK Tracking", keeping other parameters default.
4. Set "Span" to 10 MHz.
5. Adjust "RBW" to 1 kHz and "VBW" to 100 Hz to stabilize the trace.
6. THD appears in the plot's top-right corner; the harmonic table below lists frequency, amplitude, and offset from the fundamental for each harmonic.

Figure 42: Third-harmonic measurement

12. Phase Noise Measurement Mode

12.1 Version Requirements

Check the software/firmware version under Viewing Instrument Information, and confirm it meets the minimum GUI, API, FPGA, and MCU versions required for this feature; update per the Software and Firmware Update chapter if needed.

12.2 Enabling Phase Noise Measurement

Once the software and firmware meet the version requirement, restart SpecICX-gen3 and click "Mode" -> "Phase Noise".

12.3 Parameters

Parameter	Description
Center	Center frequency of the fundamental.
Start Offset	Start of the measured offset range, 1 Hz to 9 MHz.
Stop Offset	End of the measured offset range, 10 Hz to 10 MHz.
Threshold	Level above which a carrier is recognized.
RBW/Offset	RBW ratio per frequency segment, 0.01 to 0.3.
Detect	Frame detection rate; raise it near the carrier if low-frequency jitter is significant.
Average	Number of trace averages.
Smooth	Enables/disables trace smoothing.
Window Length	Smoothing window length, 0-10%.

12.4 Procedure

12.4.1 Known Carrier Information

Example: phase noise of a 1 GHz, 0 dBm signal, 100 Hz to 10 MHz offset.

1. Set "Center" to 1 GHz, "Start Offset" to 100 Hz, "Stop Offset" to 10 MHz; leave other parameters default.
2. If the signal shows strong jitter near the carrier, raise the frame detection rate under "Meas" -> "Detect" for the relevant band.

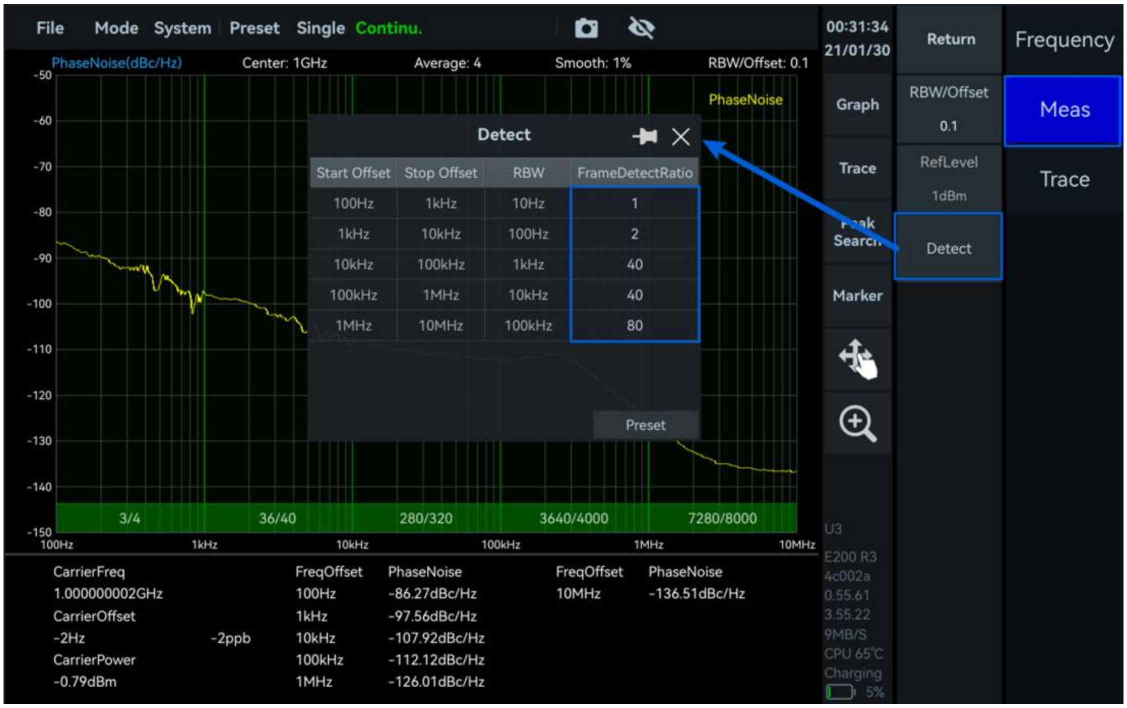

Figure 43: Frame detection settings

3. If spurious components appear in the SSB phase noise trace, gradually increase "Window Length" under "Trace" to suppress them.
4. The instrument measures phase noise across the specified offset range; results appear in the phase noise table, including carrier information and dBc/Hz values at characteristic offsets.

Figure 44: Phase noise measurement result

12.4.2 Unknown Carrier Information

1. Click "Carrier" to have the instrument scan the full band and locate a peak above the carrier threshold as the test carrier.
2. Once located, follow the Known Carrier Information procedure above to set the offset range and measure.

13. Mapping Mode

This chapter covers key Mapping mode parameters and how to use it for automated measurements that build signal-distribution heat maps.

13.1 Version Requirements

Confirm the GUI version meets the minimum required for Mapping mode under Viewing Instrument Information.

13.2 Parameters

Main Settings Area

Parameter	Description
Point Spacing	Distance in meters between recorded points; applies only in Auto mode.
Point Size	Display radius of a data point on the map.
Point Border	Adds a black outline to points for contrast against complex map backgrounds.
Point Opacity	Transparency of data-point color; lower values are more transparent.

Chart Settings Area

Control	Description
Manual	Tap the map (or click) to capture and plot a data point at the current location.
Auto	With GPS locked, automatically collects and plots points per the Point Spacing setting.
Select	Select/deselect existing points; selecting shows frequency, bandwidth, power, peak, OBW, time, and coordinates. "Direction" adds a rotatable heading indicator to the point.
Clear Points	Clears all plotted points.
Map	Imports a custom image as the base map.
Latitude and Longitude	Sets the imported image's geographic bounds for coordinate calibration.
Zoom controls	Zoom in/out on the map, or measure distance between two clicked points.

13.3 Procedure

Importing a custom image and generating a heat map via manual point marking:

1. Connect the test antenna to the instrument's RF input.
2. Click "Map" in the main settings area to enable image-map mode, and set point size, border, and opacity.
3. Set "Center" to 2.13 GHz and "Span" to 100 MHz; under "Meas", set Center Frequency to 2.13 GHz and Integration Bandwidth to 40 MHz.
4. Click "Map" in the chart settings area and import a custom image.
5. Click "Latitude and Longitude" and enter the image's geographic bounds.
6. Click "Manual", move the instrument, and tap corresponding map positions to mark points; the heat map builds from the measured power values.

Figure 45: Heat map from a custom image map

7. Click "Select" and choose a point to view its detailed parameters in the floating panel.
8. Enable "Direction" on a selected point to show and adjust its heading indicator.
9. Current channel power appears in the upper right; the full measurement table appears in the lower right (double-click to enlarge).

14. MSCAN Mode

14.1 Description

MSCAN mode sweeps a user-defined list of frequency bands sequentially, suited to fast multi-band preview and analysis. Its interface has three parts:

- Preview Diagram: composite spectrum of all configured bands, with the selected band highlighted
- Configuration List: per-band parameters — center frequency, span, reference level, dwell time, decimation, FFT size, detection count, detector type, IFAGC, XPPS trigger, IQ output, window type
- Spectrum and IQvT View: spectrum and IQ time-domain view for the selected band

14.2 General Parameters

Parameter	Description
#x Center	Center frequency of the selected band in the sweep list.
SweepTime	Total time to sweep all listed bands.
# (Index)	Band index, 1-128.
Center (Hz)	Per-band center frequency.
Span (Hz)	Per-band analysis bandwidth = 101.526 ÷ decimation factor.
Ref (dBm)	Per-band reference level, −100 to 27.
DwellTime (s)	Per-band dwell time.
Decimate	Decimation factor (reduces span), 2^n for n = 1 to 12.
FFTSize	FFT points per band, 2^n for n = 5 to 12.
DetectCount	Sweeps performed per band.
Detector	Sample, Positive Peak, Average, Negative Peak, Max Power, Raw Frame, or RMS.
IFAGC	Keeps in-band signal level within range when enabled.
XPPStrigger	On: 1PPS trigger for the band. Off: bus trigger.
IQPlayBack	Shows the band's IQ time-domain data in the IQvT view when enabled.
Window	B-Nuttall, Flat-Top, LowSideLobe, Rectangle, or Kaiser.
Insert / Delete / Import / Export	Manage list entries; Export saves to /data.

14.3 Procedure

Example Configuration

Band	Configuration
1	Center 500 MHz; Span 50 MHz; FFTSize 2048; Detector PosPeak; Window FlatTop
2	Center 2.4 GHz; Span 25 MHz; FFTSize 1024; Detector RMS; IQPlayBack On; Window Kaiser

Operating Steps

Using an ICX instrument with a 50 MHz analysis bandwidth option as the example:

1. Click "Mode" -> "MSCAN" and delete the default configuration.

2. Configure Band 1: Center = 500 MHz, Ref = 0 dBm, Decimate = 1, FFTSize = 2048, Window = FlatTop.
3. Configure Band 2: Center = 2.4 GHz, Ref = −20 dBm, Decimate = 2, FFTSize = 1024, enable IQPlayBack, Window = Kaiser.
4. The preview shows the combined spectrum with the selected band highlighted; its spectrum and IQ time-domain view (if enabled) display alongside.
5. Export the configuration list via "Export", or reload a saved one via "Import".

Figure 46: MSCAN configuration list

15. Signal/Tracking Generator Function (Option 02)

The built-in signal/tracking generator option outputs single-tone, frequency-sweep, and power-sweep signals, and supports tracking-generator operation. Available on current-generation workstation ICX instruments only.

15.1 General Parameters

Parameter	Description
RF Out	On/off control for signal source output.
Mode	Fixed Frequency, Frequency Sweep, Power Sweep, or TG.
Center (Fixed)	Frequency for single-tone and power-sweep signals.
Level (Fixed)	Power for single-tone and frequency-sweep signals.
FreqSweep: Start/Stop/Step/Dwell Time	Defines the frequency-sweep signal's range, step, and dwell time.

Parameter	Description
PowerSweep: Center/Start/Stop/Step/DwellTime	Defines the power-sweep signal's frequency, power range, step, and dwell time.
TG: Output Power	Amplitude of the tracking-generator output.
TG: Save/Show Reference	Save or recall a reference trace for comparison.
TG: Normalize	Normalizes against the reference trace to remove system path loss and isolate the DUT's response.

15.2 Using the Signal Generator

The signal is output via the instrument's RFOUT port, either feeding another device or looped back to the instrument's own RF input. The examples below use a self-transmit-and-receive setup.

15.2.1 Output a Single-Tone Signal

1. Click "AUXS" -> "RF On" to enable the signal source.
2. Set "Center" to 1 GHz and "Level" to −20 dBm.

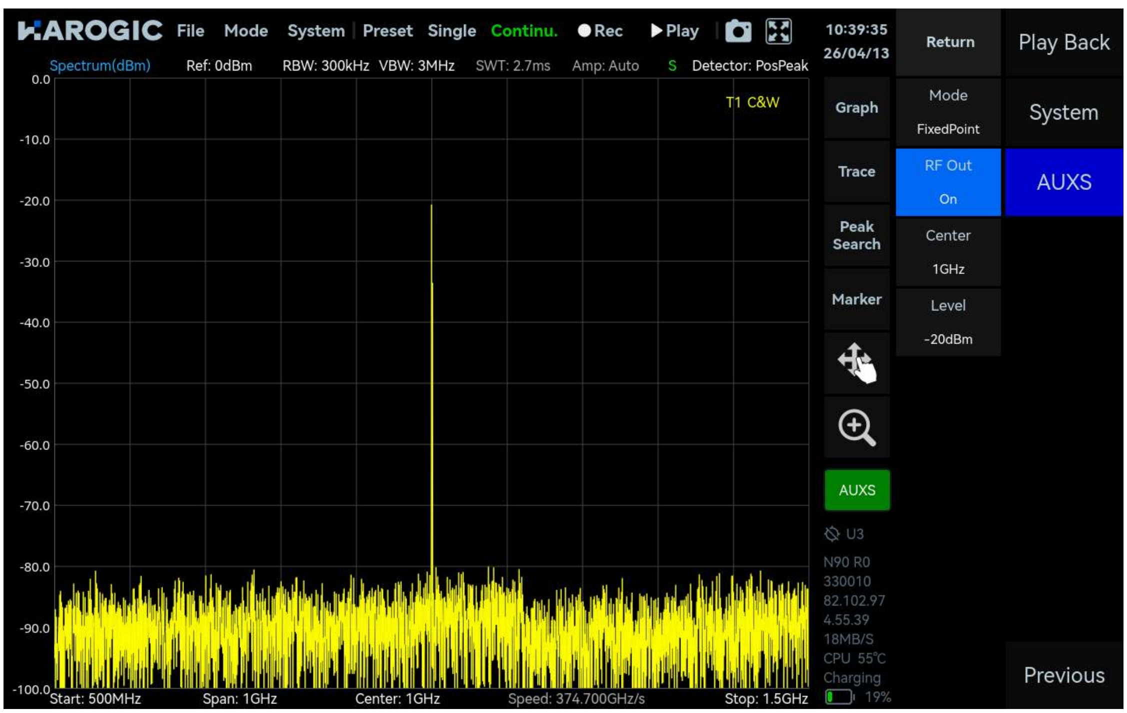

Figure 47: Single-tone signal output

15.2.2 Output a Frequency-Sweep Signal

1. Click "AUXS", set "Mode" to Frequency Sweep, and enable RF output.
2. Set Output Power = −20 dBm, Start = 3 GHz, Stop = 3.5 GHz, Step = 40 MHz, Dwell Time = 8 ms.

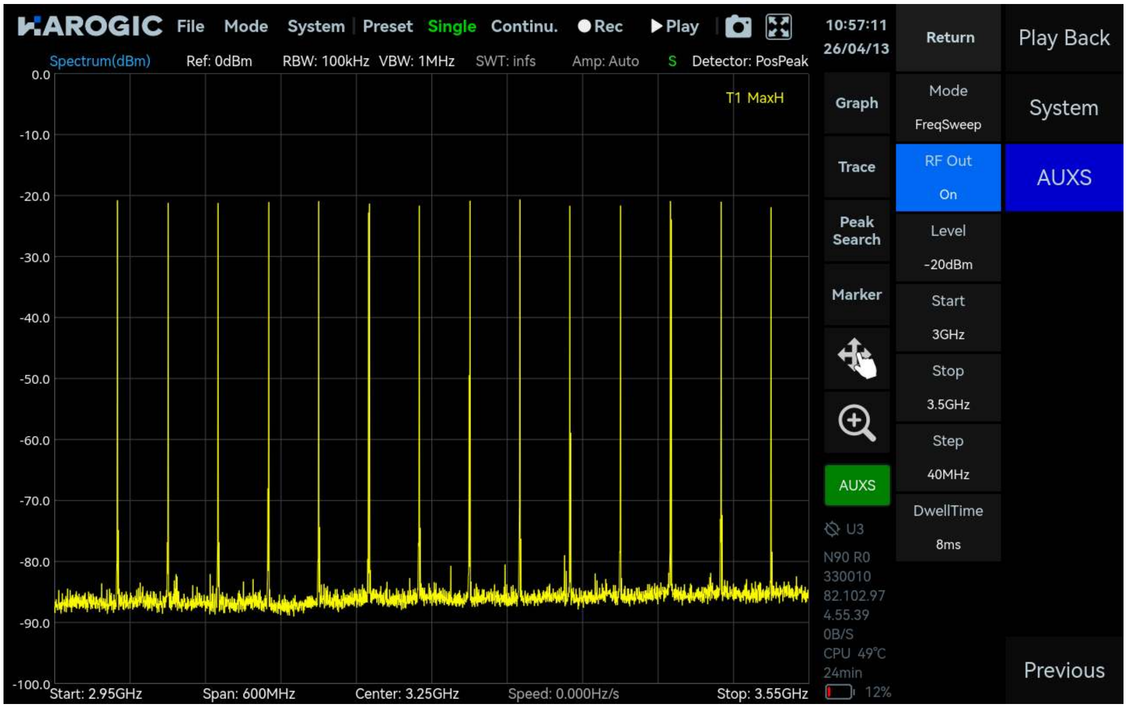

Figure 48: Frequency-sweep signal output

15.2.3 Output a Power-Sweep Signal

1. Click "AUXS", set "Mode" to Power Sweep, and enable RF output.
2. Set Center = 1 GHz, Start Power = −40 dBm, Stop Power = −10 dBm, Step = 1 dBm, Dwell Time = 100 ms.

15.2.4 Tracking Generator (TG)

TG mode synchronizes the RF output to the current sweep frequency, for measuring a device under test's (DUT) frequency response.

1. Set Center Frequency and Span to the DUT's operating band.
2. Click "AUXS" -> "RF On", select TG mode, and set an appropriate output power for the DUT.
3. Connect the TG output directly to the RF input with a high-quality coaxial cable.
4. Once the trace stabilizes, press "Through Calibration" to capture the cable-loss baseline, then enable normalization (the trace should flatten to roughly 0 dB).
5. Insert the DUT between the TG output and RF input; the displayed trace now shows the DUT's true frequency response.

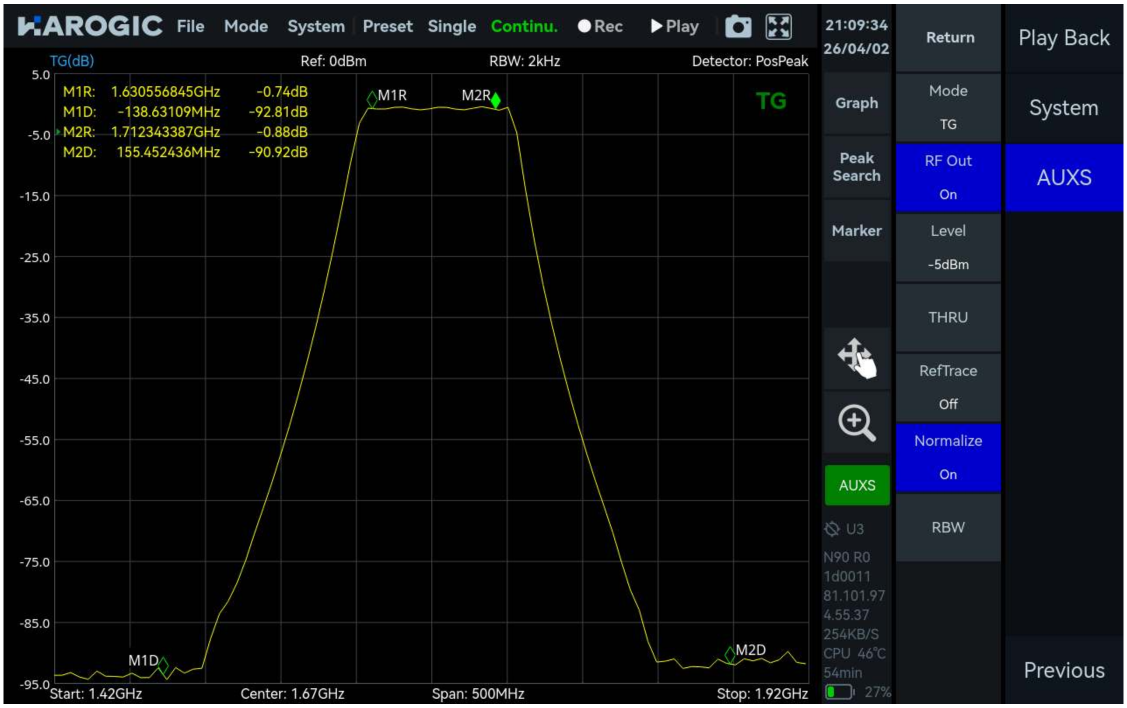

Figure 49: Frequency response of a band-pass filter

16. Additional Functions

This chapter briefly covers IF output, triggering, external reference clock input, and multi-instrument operation.

16.1 Trigger Functions

16.1.1 SWP Frequency Sweep Mode

Parameter	Description
TriggerSource	FreeRun, Ext.PerHop, Ext.PerSweep, Ext.PerProfile, InXpps.PerHop, InXpps.PerSweep, InXpps.PerProfile.
TriggerEdge	RisingEdge, FallingEdge, or DoubleEdge.
TriggerOut	None, PerHop (each analysis frame), PerSweep (each trace scan), or PerProfile (each config switch).
PulsePolarity	Positive or Negative.

16.1.2 Fixed Frequency Point Mode (IQS, DET, RTA)

Parameter	Description
TriggerSource	External, Bus, Level, Timer, DevSyncByExt, DevSyncBy1PPS, or GNSS1PPS.
TriggerEdge	RisingEdge, FallingEdge, or DoubleEdge.
TriggerDelay / PreTrigger	Delay acquisition after trigger, or capture data before it.

Parameter	Description
ReTrigger – Count / Period	Number of additional responses per trigger, and the interval between them.
Level / SafeTime (Level trigger)	Trigger threshold and debounce time.
Period / Sync (Timer trigger)	Trigger period, and synchronization to an external trigger edge or GNSS-1PPS edge (continuous or single-shot).

16.2 IF Output Application Guide

The analog IF output signal runs at 312.5 MHz ± 50 MHz. Each instrument's specific IF center frequency is recorded in its IF calibration file, found in the /bin/CalFile folder of the software install directory.

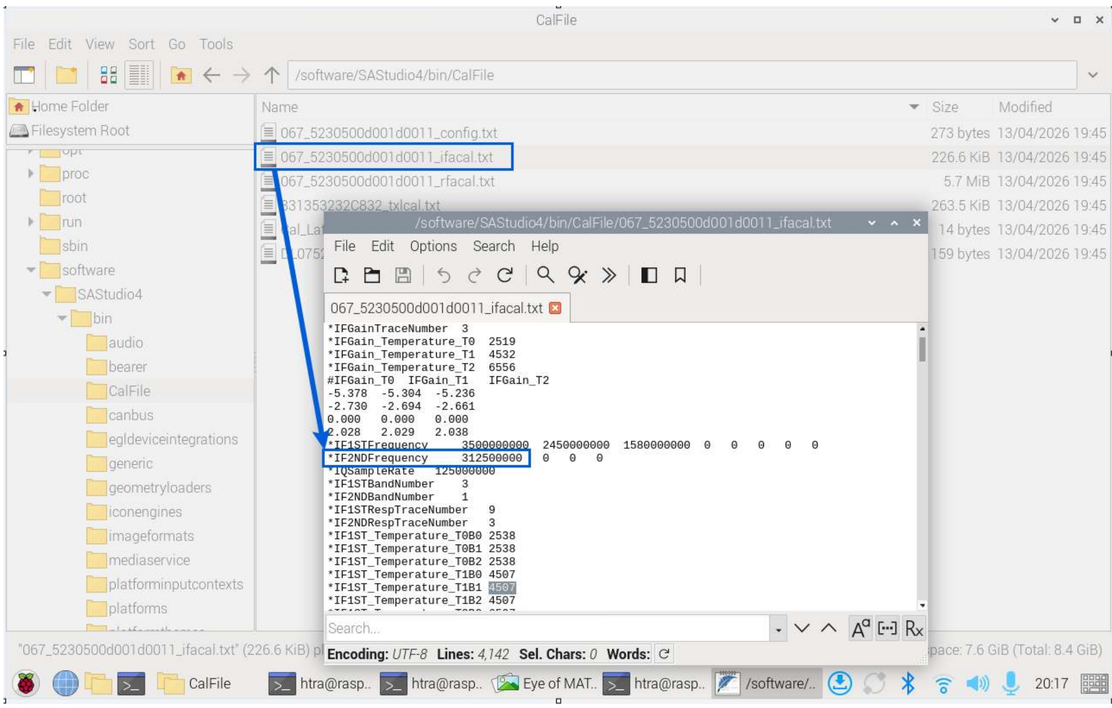

Figure 50: Checking the IF output center frequency

16.3 External Reference Clock Input

1. Connect the external reference clock per the ICX Quick Start Guide.
2. In the main setup area, click "Next" -> "System", set "RefCLKFreq" to 10 MHz, and set "RefCLKSource" to "External". A successful switch shows "External" as the active source; if it reverts to "Internal" with an error, click "Preset" to fall back to the internal clock.

Figure 51: Using an external 10 MHz reference clock

16.4 Connecting and Operating Multiple Instruments Simultaneously

A single SpecICX-gen3 window controls one instrument; running multiple windows lets you control multiple instruments concurrently, each assigned a distinct device number. Example with two workstation instruments:

1. Connect both instruments to the host computer.
2. Copy each instrument's calibration file into /bin/CalFile in the software directory.
3. Launch the software to bring up the instrument with device number 0.
4. In the software's configuration folder, open Setting.ini and change "DeviceNum=0" to "DeviceNum=1".
5. Launch the software again in a new window to bring up the instrument with device number 1.

Note: If only one instrument will be used going forward, reset "DeviceNum" to 0 in Setting.ini, or the software may fail to recognize the instrument correctly.

Figure 52: Operating two instruments simultaneously

17. Application for Options

17.1 Pulse Detection Option

17.1.1 Applying for a License

1. Check the software/firmware version under Viewing Instrument Information.
2. Confirm the version meets the minimum required for this option; update per the Software and Firmware Update chapter if needed.
3. Click "System" -> "About", screenshot the full interface, and send it to BNC support to request the license.

17.1.2 Installing the License

Workstation ICX instruments:

1. Copy the received license file into /bin/CalFile in the software directory.
2. Restart the software; click "Mode" -> "Power Detection" and enable "Pulse Det" in the main settings area.

Handheld ICX instruments:

3. Extract the received "Option" folder onto a USB drive.
4. Click "File" -> "Exit" to close the software.
5. Insert the USB drive; confirm the "removable medium inserted" prompt.
6. Open the "Option" folder and run the installer icon; wait for the success confirmation.

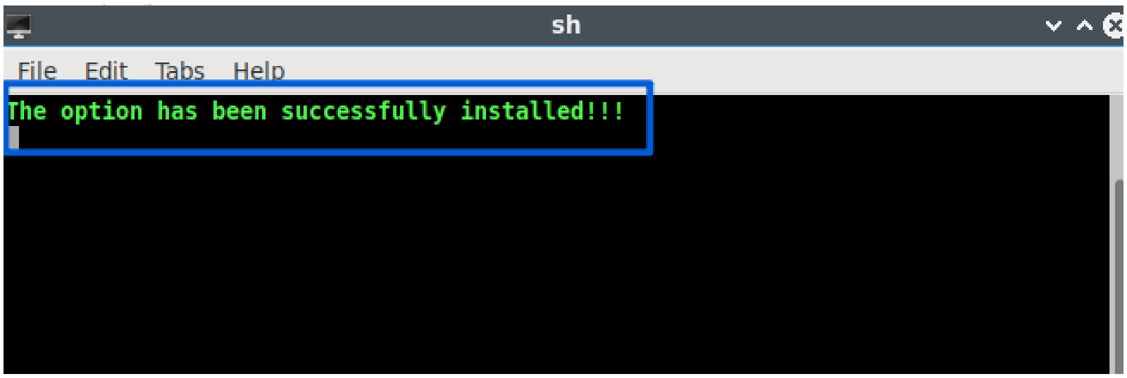

Figure 53: Installing the Pulse Detection option license

7. Close the prompt, restart the software, click "Mode" -> "Pulse Detection", and enable "Pulse Det".

17.2 Digital Demodulation Option

17.2.1 Applying for a License and Library

Follow the same license-request process as the Pulse Detection option to obtain the license and demodulation library.

17.2.2 Installing the License and Library

Workstation ICX instruments:

1. Copy the demodulation library file into /bin in the software directory.
2. Copy the license file into /bin/CalFile in the software directory.
3. Restart the software; click "Mode" -> "Digital Demod".

Handheld ICX instruments:

4. Follow the Pulse Detection installer steps above, using the installer icon to install both the license and library.
5. Close the prompt, start the software, and click "Mode" -> "Digital Demod".

17.2.3 Signal/Tracking Generator Option

Adding the signal/tracking generator option to an existing instrument requires returning it to BNC for a hardware upgrade.

18. Software and Firmware Update

This chapter covers updating SpecICX-gen3's software, FPGA, MCU, and bus versions.

18.1 Version Requirements

Check current versions under Viewing Instrument Information; if the software reports that an update can't be performed, contact BNC support.

18.2 Update Concepts

- The update dialog shows current and target versions for software, FPGA, MCU, and bus.
- Update Content lists what's changed in the target version.
- Update Notification controls whether a pop-up appears automatically when a new version is detected.

Update Method	Description
Online	Downloads and installs the latest release directly from the update server.
Default	Updates from locally stored default update files.
Local	Manually select a local update package.

18.3 Online Update

1. Click "System" -> "Update".
2. Enable "Update Notification" with the device connected to the internet; a pop-up appears automatically when a new version is detected, or open the Update window manually via the same menu path.
3. Set "Update Method" to Online; once the package downloads and parses, release notes appear and the "Update" button becomes active.
4. Review the current vs. target version details and the changelog, then click "Update".

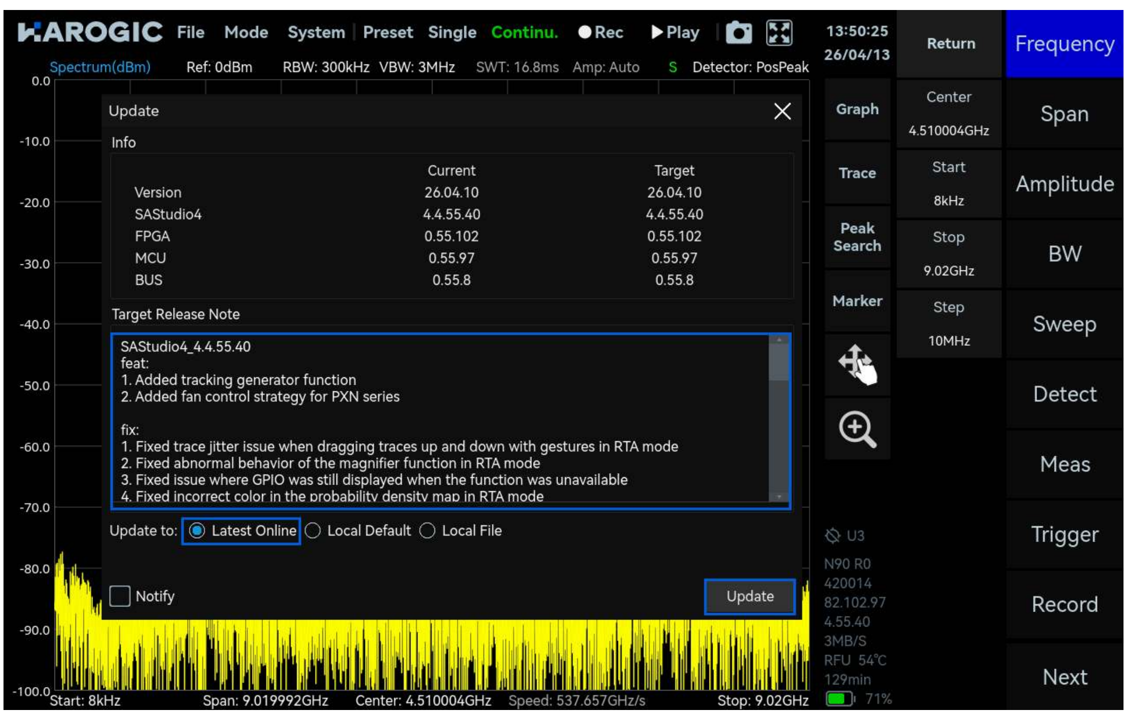

Figure 54: Updating software and firmware online

5. The software exits and enters the upgrade process; keep the update window open until it completes and the software restarts automatically.

6. Close the update window, then check "System" -> "About" to confirm the new firmware version.

18.4 Offline Update

Download the current software release from BNC's support portal for your instrument series (workstation or handheld). Using a handheld instrument as the example:

1. Copy the downloaded installation package to a USB drive; insert it into the instrument and confirm the "removable medium inserted" prompt.
2. Click "System" -> "Update".
3. Select "Local", click "Open" to choose the package, then click "Update".

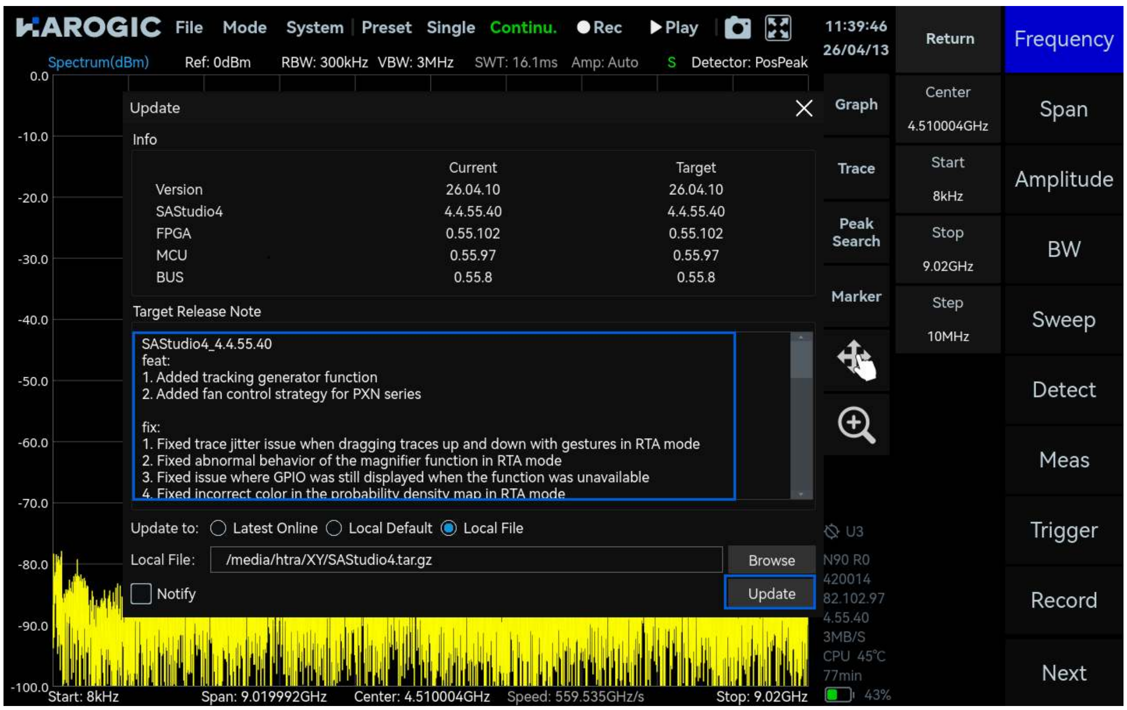

Figure 55: Offline software and firmware update

4. The software exits and enters the upgrade process; keep the update window open until it completes and the software restarts automatically.

Appendix

The reference tables in this appendix describe the on-disk record file layout for each working mode. This layout is inherited from the underlying measurement engine shared across the ICX platform; engineering should confirm byte-for-byte parity with the SpecICX-gen3 build before any customer-facing parsing tools are built against it.

Appendix 1: Record File Format Description

1.1 File Naming Format

Record file names follow: last four digits of device ID, start date/time (yyyymmdd_hhmmss), part index, and suffix — for example, 0028_yyyymmdd_hhmmss_partx.suffix. All sub-files from the same recording session share the same date-time stamp, reflecting when the recording began.

1.2 Conversion of Structures into Byte Arrays

The structures SWP_Profile_TypeDef, SWP_TraceInfo_TypeDef, IQS_Profile_TypeDef, IQS_StreamInfo_TypeDef, DET_Profile_TypeDef, DET_StreamInfo_TypeDef, RTA_Profile_TypeDef, and RTA_FrameInfo_TypeDef are serialized to byte arrays using msgpack, and must be decoded with msgpack when reading recorded files.

Appendix 2: Standard Spectrum Analysis Mode

SWP record files use a custom ".spectrum" format.

File Structure Overview

Section	Contents
Byte 0 to (71 + 10M)	File information and packet index (1 MB = 1024 × 1024 bytes).
Byte (72 + 10M) to (73 + 10M + length)	SWP_Profile_TypeDef (current config), SWP_Profile_TypeDef (default config), and SWP_TraceInfo_TypeDef; "length" is the combined byte size of these three structures.
Byte (74 + 10M + length) to end of file	Per-packet byte length followed by SWP data packets, each containing frequency/power arrays, HopIndex, FrameIndex, and MeasAuxInfo_TypeDef.

File Header

Field Name	Data Type	Bytes	Notes
File ID	uint16_t	2	Big Endian
Protocol markers	uint8_t ×4	4	Fixed values 0x8c, 0x22, 0x52, 0x9b — Big Endian
Protocol Version	uint8_t ×2	2	Big Endian
API Version Information	uint32_t	4	Big Endian
Reserved	—	56	—
Number of Data Packets in Current File	uint64_t	8	Big Endian
Packet Index	QList	10M	Big Endian

Configuration / Default Configuration / Trace Info (msgpack)

Configuration Information length field precedes the block (uint16_t, Big Endian). Fields in SWP_Profile_TypeDef (config and default config share the same layout):

Field Name	Data Type	Bytes	Notes
StartFreq_Hz / StopFreq_Hz / CenterFreq_Hz / Span_Hz / RefLevel_dBm	double	8 each	—
RBW_Hz / VBW_Hz / SweepTime / TraceBinSize_Hz	double	8 each	—
FreqAssignment / Window / RBWMode / VBWMode / SweepTimeMode	int	4 each	—
Detector / TraceFormat / TraceDetectMode / TraceDetector	int	4 each	—
TracePoints	uint32_t	4	—
TracePointsStrategy / TraceAlign / FFTExecutionStrategy / RxPort / SpurRejection	int	4 each	—
ReferenceClockSource	int	4	—
ReferenceClockFrequency	double	8	—
EnableReferenceClockOut	uint8_t	1	—
SystemClockSource	int	4	—
ExternalSystemClockFrequency	double	8	—
TriggerSource / TriggerEdge / TriggerOutMode / TriggerOutPulsePolarity	int	4 each	—
PowerBalance	uint32_t	4	—
GainStrategy / Preamplifier	int	4 each	—
AnalogIFBWGrade / IFGainGrade / EnableDebugMode / CalibrationSettings	uint8_t	1 each	—
Atten	int8_t	1	—
TraceType / LOOptimization	int	4 each	—
Field Name	**Data Type**	**Bytes**	**Notes**
FullsweepTracePoints / PartialsweepTracePoints / TotalHops	int	4 each	Trace Info structure
UserStartIndex / UserStopIndex	uint32_t	4 each	—

Field Name	Data Type	Bytes	Notes
TraceBinBW_Hz / StartFreq_Hz / AnalysisBW_Hz	double	8 each	—
TraceDetectRatio / DecimateFactor	int	4 each	—
FrameTimeMultiple	float	4	—
FrameTime / EstimateMinSweepTime	double	8 each	—
DataFormat	int	4	—
SamplePoints	uint64_t	8	—
GainParameter	uint32_t	4	—
DSPPlatform	int	4	—

Data Packet (repeats for each packet)

Field Name	Data Type	Bytes	Notes
Data Packet Length	int	4	Big Endian
Frequency Array	double × N	8×N	N = PartialsweepTracePoints; platform-dependent endianness
Power Array	float × N	4×N	Platform-dependent endianness
HopIndex / FrameIndex	int	4 each	Big Endian
MaxIndex	uint32_t	4	Big Endian
MaxPower_dBm	float	4	Big Endian
Temperature	int16_t	2	Big Endian
RFState / BBState / GainPattern	uint16_t	2 each	Big Endian
ConvertPattern	uint32_t	4	Big Endian
SysTimeStamp / AbsoluteTimeStamp	double	8 each	Big Endian
Latitude / Longitude	float	4 each	Big Endian

Appendix 3: Receiver/IQ Stream Mode

IQS record files use the standard ".wav" container format.

Chunk Overview

Chunk	Description
RIFF	Container header; file format type "WAVE".
fmt	Standard WAV format chunk.

Chunk	Description
prof	Stores IQS_Profile_TypeDef, IQS_StreamInfo_TypeDef, and DeviceInfo_TypeDef.
trig	Stores IQS_TriggerInfo_TypeDef, DeviceState_TypeDef, IQS_ScaleToV, MaxPower_dBm, and MaxIndex per IQ data packet, one-to-one with the data chunk.
data	IQ data packets in sequence.

RIFF / fmt Header

Field Name	Data Type	Bytes	Notes
Document Identifier	—	4	"RIFF" — Little Endian
Data Length	uint32_t	4	Chunk length — Little Endian
File Format Type	—	4	"WAVE"
Format Chunk Identifier	—	4	"fmt " — Little Endian
Chunk Length	uint32_t	4	16 — Little Endian
Audio Format Code	uint16_t	2	1 — Little Endian
Number of Channels	uint16_t	2	2 — Little Endian
Sampling Frequency	uint32_t	4	Little Endian
Byte Rate	uint32_t	4	Little Endian
Block Align	uint16_t	2	Little Endian
Bits per Sample	uint16_t	2	Little Endian

prof Chunk — IQS_Profile_TypeDef

Field Name	Data Type	Bytes	Notes
Format Chunk Identifier / Chunk Length	— / uint32_t	4 / 4	"prof" — Little Endian
File ID	uint16_t	2	Big Endian
Protocol markers	uint8_t ×4	4	0x8c, 0x22, 0x52, 0x9b — Big Endian
Protocol Version	uint8_t ×2	2	Big Endian
API Version Information	uint32_t	4	Big Endian
Reserved	—	56	—
IQS_Profile + IQS_StreamInfo Structure Byte Length	uint16_t	2	Big Endian
CenterFreq_Hz / RefLevel_dBm	double	8 each	—
DecimateFactor	uint32_t	4	—
RxPort	int	4	—
BusTimeout_ms	uint32_t	4	—

Field Name	Data Type	Bytes	Notes
TriggerSource / TriggerEdge / TriggerMode	int	4 each	—
TriggerLength	uint64_t	8	—
TriggerOutMode / TriggerOutPulsePolarity	int	4 each	—
TriggerLevel_dBm / TriggerLevel_SafeTime / TriggerDelay / PreTriggerTime	double	8 each	—
TriggerTimerSync	int	4	—
TriggerTimer_Period	double	8	—
EnableReTrigger	uint8_t	1	—
ReTrigger_Period	double	8	—
ReTrigger_Count	uint16_t	2	—
DataFormat / GainStrategy / Preamplifier	int	4 each	—
AnalogIFBWGrade / IFGainGrade / EnableDebugMode	uint8_t	1 each	—
ReferenceClockSource	int	4	—
ReferenceClockFrequency	double	8	—
EnableReferenceClockOut	uint8_t	1	—
SystemClockSource	int	4	—
ExternalSystemClockFrequency / NativeIQSampleRate_SPS	double	8 each	—
EnableIFAGC	uint8_t	1	—
Atten	int8_t	1	—
DCCancelerMode / QDCMode	int	4 each	—
QDCIGain / QDCQGain / QDCPhaseComp	float	4 each	—
DCCIOffset / DCCQOffset	int8_t	1 each	—
LOOptimization	int	4	—

IQS_StreamInfo / DeviceInfo

Field Name	Data Type	Bytes	Notes
Bandwidth / IQSampleRate	double	8 each	—
PacketCount / StreamSamples / StreamDataSize	uint64_t	8 each	—

Field Name	Data Type	Bytes	Notes
PacketSamples / PacketDataSize / GainParameter	uint32_t	4 each	—
DeviceInfo Structure Byte Length	uint16_t	2	Big Endian
DeviceUID	uint64_t	8	Big Endian
Model / HardwareVersion	uint16_t	2 each	Big Endian
MFWVersion / FFWVersion	uint32_t	4 each	Big Endian

trig Chunk — IQS_TriggerInfo / DeviceState

Field Name	Data Type	Bytes	Notes
Chunk Identifier / Chunk Length	— / uint32_t	4 / 4	"trig" — Little Endian
IQS_TriggerInfo Structure Byte Length	uint16_t	2	Big Endian
SysTimerCountOfFirstDataPoint	uint64_t	8	Big Endian
InPacketTriggeredDataSize / InPacketTriggerEdges	uint16_t	2 each	Big Endian
StartDataIndexOfTriggerEdges[25]	uint32_t ×25	100	Platform-dependent
SysTimerCountOfEdges[25]	uint64_t ×25	200	Platform-dependent
EdgeType[25]	int8_t ×25	25	Platform-dependent
DeviceState Structure Byte Length	uint16_t	2	Big Endian
Temperature	int16_t	2	Big Endian
RFState / BBState / GainPattern	uint16_t	2 each	Big Endian
AbsoluteTimeStamp	double	8	Big Endian
Latitude / Longitude	float	4 each	Big Endian
RFCFreq	int64_t	8	Big Endian
ConvertPattern / NCOFTW / SampleRate	uint32_t	4 each	Big Endian
CPU_BCFlag / IFOverflow / DecimateFactor / OptionState	uint16_t	2 each	Big Endian
IQS_ScaleToV / MaxPower_dBm	float	4 each	Big Endian
MaxIndex	uint32_t	4	Big Endian

data Chunk

Field Name	Data Type	Bytes	Notes
Chunk Identifier	—	4	"data" (at offset 25×1024×1024+400)
Chunk Length	uint32_t	4	Little Endian
IQ Data Packet (repeats)	—	64968 per packet	Platform-dependent; packet count = Chunk Length ÷ 64968

Appendix 4: Power Detection Mode

DET record files use the standard ".wav" container format, though they are not directly playable by third-party audio software.

Chunk Overview

Chunk	Description
RIFF / fmt	Standard WAV header, matching Appendix 3.
prof	Stores DET_Profile_TypeDef and DET_StreamInfo_TypeDef.
trig	Stores IQS_TriggerInfo_TypeDef and MeasAuxInfo_TypeDef, one-to-one with the data chunk.
data	DET data packets in sequence.

prof Chunk — DET_Profile_TypeDef

Field Name	Data Type	Bytes	Notes
File ID	uint16_t	2	Big Endian
Protocol markers	uint8_t ×4	4	0x8c, 0x22, 0x52, 0x9b — Big Endian
Protocol Version	uint8_t ×2	2	Big Endian
API Version Information	uint32_t	4	Big Endian
DET_Profile + DET_StreamInfo Structure Byte Length	uint16_t	2	Big Endian
CenterFreq_Hz / RefLevel_dBm	double	8 each	—
DecimateFactor	uint32_t	4	—
RxPort	int	4	—
BusTimeout_ms	uint32_t	4	—
TriggerSource / TriggerEdge / TriggerMode	int	4 each	—
TriggerLength	uint64_t	8	—
TriggerOutMode / TriggerOutPulsePolarity	int	4 each	—

Field Name	Data Type	Bytes	Notes
TriggerLevel_dBm / TriggerLevel_SafeTime / TriggerDelay / PreTriggerTime	double	8 each	—
TriggerTimerSync	int	4	—
TriggerTimer_Period	double	8	—
EnableReTrigger	uint8_t	1	—
ReTrigger_Period	double	8	—
ReTrigger_Count	uint16_t	2	—
Detector / DetectRatio	int / uint16_t	4 / 2	—
GainStrategy / Preamplifier	int	4 each	—
AnalogIFBWGrade / IFGainGrade / EnableDebugMode	uint8_t	1 each	—
ReferenceClockSource	int	4	—
ReferenceClockFrequency	double	8	—
EnableReferenceClockOut	uint8_t	1	—
SystemClockSource	int	4	—
ExternalSystemClockFrequency	double	8	—
Atten	int8_t	1	—
DCCancelerMode / QDCMode	int	4 each	—
QDCIGain / QDCQGain / QDCPhaseComp	float	4 each	—
DCCIOffset / DCCQOffset	int8_t	1 each	—
LOOptimization	int	4	—

DET_StreamInfo

Field Name	Data Type	Bytes	Notes
PacketCount / StreamSamples / StreamDataSize	uint64_t	8 each	—
PacketSamples / PacketDataSize	uint32_t	4 each	—
TimeResolution	double	8	—
GainParameter	uint32_t	4	—

trig Chunk — IQS_TriggerInfo / MeasAuxInfo

Field Name	Data Type	Bytes	Notes
Format Chunk Identifier / Chunk Length	— / uint32_t	4 / 4	"trig" — Little Endian
IQS_TriggerInfo Structure Byte Length	uint16_t	2	Big Endian
SysTimerCountOfFirstDataPoint	uint64_t	8	Big Endian
InPacketTriggeredDataSize / InPacketTriggerEdges	uint16_t	2 each	Big Endian
StartDataIndexOfTriggerEdges[25]	uint32_t ×25	100	Platform-dependent
SysTimerCountOfEdges[25]	uint64_t ×25	200	Platform-dependent
EdgeType[25]	int8_t ×25	25	Platform-dependent
MeasAuxInfo Structure Byte Length	uint16_t	2	Big Endian
MaxIndex	uint32_t	4	Big Endian
MaxPower_dBm	float	4	Big Endian
Temperature	int16_t	2	Big Endian
RFState / BBState / GainPattern	uint16_t	2 each	Big Endian
ConvertPattern	uint32_t	4	Big Endian
SysTimeStamp / AbsoluteTimeStamp	double	8 each	Big Endian
Latitude / Longitude / ScaleToV	float	4 each	Big Endian

data Chunk

Field Name	Data Type	Bytes	Notes
Chunk Identifier	—	4	"data" (at offset 25×1024×1024+400)
Chunk Length	uint32_t	4	Little Endian
DET Data Packet (repeats)	—	64968 per packet	Platform-dependent; packet count = Chunk Length ÷ 64968

Appendix 5: Real-Time Spectrum Analysis Mode

RTA record files use a custom ".rtspectrum" format.

File Structure Overview

Section	Contents
Byte 0 to (71 + 10M)	File information and packet index.

Section	Contents
Byte (72 + 10M) to (73 + 10M + length)	RTA_Profile_TypeDef and RTA_FrameInfo_TypeDef; "length" is their combined byte size.
Byte (74 + 10M + length) to end of file	Per-packet byte length followed by RTA data packets, each containing SpectrumStream, RTA_PlotInfo_TypeDef, RTA_TriggerInfo_TypeDef, and MeasAuxInfo_TypeDef.

File Header

Field Name	Data Type	Bytes	Notes
File ID	uint16_t	2	Big Endian
Protocol markers	uint8_t ×4	4	0x8c, 0x22, 0x52, 0x9b — Big Endian
Protocol Version	uint8_t ×2	2	Big Endian
API Version Information	uint32_t	4	Big Endian
Number of Data Packets in Current File	uint64_t	8	Big Endian
Packet Index	QList	10M	Big Endian

Configuration Information (RTA_Profile_TypeDef)

Field Name	Data Type	Bytes	Notes
Configuration Information + Trace Information Length	uint16_t	2	Big Endian
CenterFreq_Hz / RefLevel_dBm / RBW_Hz / VBW_Hz	double	8 each	—
RBWMode / VBWMode / DecimateFactor	int / int / uint32_t	4 each	—
Window / SweepTimeMode	int	4 each	—
SweepTime	double	8	—
Detector / TraceDetectMode	int	4 each	—
TraceDetectRatio	uint32_t	4	—
TraceDetector / RxPort	int	4 each	—
BusTimeout_ms	uint32_t	4	—
TriggerSource / TriggerEdge / TriggerMode	int	4 each	—
TriggerAcqTime	double	8	—
TriggerOutMode / TriggerOutPulsePolarity	int	4 each	—

Field Name	Data Type	Bytes	Notes
TriggerLevel_dBm / TriggerLevel_SafeTime / TriggerDelay / PreTriggerTime	double	8 each	—
TriggerTimerSync	int	4	—
TriggerTimer_Period	double	8	—
EnableReTrigger	uint8_t	1	—
ReTrigger_Period	double	8	—
ReTrigger_Count	uint16_t	2	—
GainStrategy / Preamplifier	int	4 each	—
AnalogIFBWGrade / IFGainGrade / EnableDebugMode	uint8_t	1 each	—
ReferenceClockSource	int	4	—
ReferenceClockFrequency	double	8	—
EnableReferenceClockOut	uint8_t	1	—
SystemClockSource	int	4	—
ExternalSystemClockFrequency	double	8	—
Atten	int8_t	1	—
DCCancelerMode / QDCMode	int	4 each	—
QDCIGain / QDCQGain / QDCPhaseComp	float	4 each	—
DCCIOffset / DCCQOffset	int8_t	1 each	—
LOOptimization	int	4	—

Trace Information (RTA_FrameInfo_TypeDef)

Field Name	Data Type	Bytes	Notes
StartFrequency_Hz / StopFrequency_Hz	double	8 each	—
POI / TraceTimestampStep / TimeResolution / PacketAcqTime	double	8 each	—
PacketCount / PacketFrame / FFTSize	uint32_t	4 each	—
FrameWidth / FrameHeight	uint32_t	4 each	—
PacketSamplePoints / PacketValidPoints / MaxDensityValue / GainParameter	uint32_t	4 each	—

Data Packet (repeats for each packet)

Field Name	Data Type	Bytes	Notes
Data Packet Length	int	4	Big Endian
SpectrumStream Array	uint8_t × N	N	Platform-dependent
ScaleTodBm / OffsetTodBm	float	4 each	Big Endian
SpectrumBitmapIndex / SysTimerCountOfFirstDataPoint	uint64_t	8 each	Big Endian
InPacketTriggeredDataSize / InPacketTriggerEdges	uint16_t	2 each	Big Endian
StartDataIndexOfTriggerEdges[25]	uint32_t ×25	100	Platform-dependent
SysTimerCountOfEdges[25]	uint64_t ×25	200	Platform-dependent
EdgeType[25]	int8_t ×25	25	Platform-dependent
MaxIndex	uint32_t	4	Big Endian
MaxPower_dBm	float	4	Big Endian
Temperature	int16_t	2	Big Endian
RFState / BBState / GainPattern	uint16_t	2 each	Big Endian
ConvertPattern	uint32_t	4	Big Endian
SysTimeStamp / AbsoluteTimeStamp	double	8 each	Big Endian
Latitude / Longitude	float	4 each	Big Endian